

The resource elasticity of control

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This study makes the **valuable** claim that people track, specifically, the elasticity of control (that is, the degree to which outcome depends on how many resources - such as money - are invested), and that control elasticity is impaired in certain types of psychopathology. A novel task is introduced that provides **solid** evidence that this learning process occurs and that human behavior is sensitive to changes in the elasticity of control. Evidence that elasticity inference is distinct from more general learning mechanisms and is related to psychopathology remains **incomplete**.

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Abstract

The ability to determine how much the environment can be controlled through our actions has long been viewed as fundamental to adaptive behavior. While traditional accounts treat controllability as a fixed property of the environment, we argue that real-world controllability often depends on the effort, time and money we are able and willing to invest. In such cases, controllability can be said to be *elastic* to invested resources. Here we propose that inferring this elasticity is essential for efficient resource allocation, and thus, elasticity misestimations result in maladaptive behavior. To test this hypothesis, we developed a novel treasure hunt game where participants encountered environments with varying degrees of controllability and elasticity. Across two pre-registered studies (N=514), we first demonstrate that people infer elasticity and adapt their resource allocation accordingly. We then present a computational model that explains how people make this inference, and identify individual elasticity biases that lead to suboptimal resource allocation. Finally, we show that overestimation of elasticity is associated with elevated psychopathology involving an impaired sense of control. These findings establish the elasticity of control as a distinct cognitive construct guiding adaptive behavior, and a computational marker for control-related maladaptive behavior.

Introduction

Taking action requires us to invest resources, such as time, money, and effort, and is thus only sensible if our actions have the potential to change the environment to our advantage. Accurate estimation of this potential – that is, of the environment’s controllability^{1–4} – enables us to make informed decisions as to whether to engage in action^{5–8}. Misestimations of controllability thus result in maladaptive behavior, and contribute to mental health disorders such as depression, anxiety, and obsessive-compulsive disorder^{9–13}.

The degree of control we possess over our environment, however, may itself depend on the resources we are willing and able to invest. For example, the control a biker has over their commute time depends on the power they are willing and able to invest in pedaling. In this respect, a highly trained biker would typically have more control than a novice. Likewise, the control a diner in a restaurant has over their meal may depend on how much money they have to spend. In such situations, controllability is not fixed but rather *elastic* to available resources (i.e., in the same sense that supply and demand may be elastic to changing prices¹⁴).

To formalize how elasticity relates to control, we build on an established definition of controllability as the fraction of reward that is controllably achievable¹⁵, χ . Uncertainty about this fraction could result from uncertainty about the amount of resources that the agent is able and willing to invest, max C. Elasticity can thus be defined as the amount of information obtained about controllability by knowing the amount of available resources: $I(\chi; \max C)$.

While only controllable environments can be elastic, the inverse is not necessarily true – controllability can be high, yet inelastic to invested resources – for example, choosing between bus routes affords equal control over commute time to anyone who can afford the basic fare (**Figure 1**; Supplementary Note 1). That said, since all actions require some resource investment, no controllable environment is completely inelastic when considering the full spectrum of possible agents, including those with insufficient resources to act (e.g., those unable to purchase a bus fare or pay for a fixed-price meal). Even in this case, however, controllability can be elastic to varying degrees.

Knowing the elasticity of control is necessary for deciding whether and how to act. Failing to achieve one’s goal, for example, should only lead one to invest more resources in an elastic environment. Critically, in many real-life domains (e.g., studying for an exam), we must use our experiences to infer the degree to which we can improve our control by investing more resources^{16–22}. Failure to do so is bound to result in maladaptive behavior, and may thus contribute to mental health disorders. Depressed patients, for instance, are less willing to expend their resources to obtain control^{23,28–29}, which could reflect an underestimation of elasticity.

Conversely, the repetitive and unusual amount of effort invested by people with obsessive-compulsive disorder in attempts to exert control^{25,26} could indicate an overestimation of elasticity, that is, a belief that adequate control can only be achieved through excessive and repeated resource investment²⁶. Ultimately, not having a good grasp of how additional efforts influence the success of one’s actions is bound to undermine one’s sense of agency^{16–17}.

Thus, here we ask whether and how people infer the elasticity of control over their environment. We hypothesize that individual differences in elasticity inference lead to a mismanagement of one’s resources when attempting to control the environment, and are thus associated with psychopathologies involving a distorted sense of control^{23–26}.

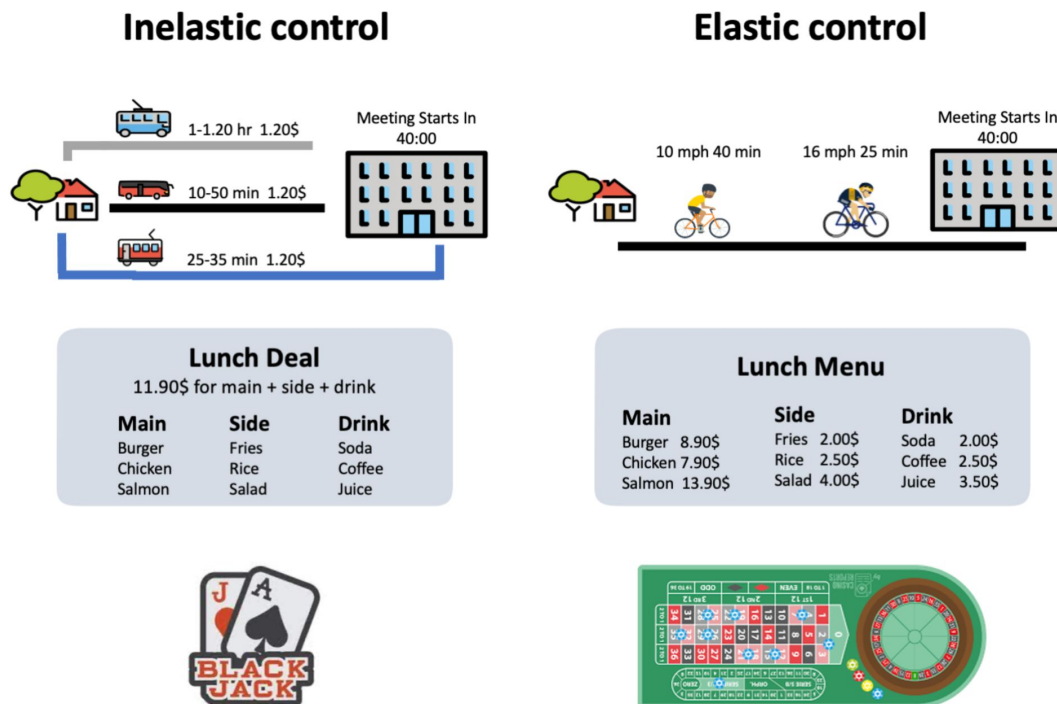


Figure 1.

Examples of inelastic vs elastic control: Top –

Choosing one of three equally priced public transport routes provides inelastic control over commute time, as control does not increase with additional available resources beyond the basic fare. In contrast, the control of a biker over commute time is elastic to the power they can invest in pedaling. **Middle** – A diner has either inelastic or elastic control over their lunch depending on whether the restaurant offers a fixed-price lunch deal where all diners have the same options, or a standard menu where more expensive dishes are only available to those able to spend more. **Bottom** – the probability of winning in blackjack is inelastic to money, as it only depends on whether one hits or stands, whereas when playing roulette, one can increase the probability of winning by investing more coins to cover more possible outcomes. Importantly, most real-world scenarios lie on a spectrum between these illustrative extremes, such that controllability is partly elastic and partly inelastic. In all situations, agents must infer the degree to which controllability is elastic to be able to determine whether the potential gains in control outweigh the costs of investing additional resources (e.g., physical exertion, money spent, time invested).

Experimental paradigms to date have conflated overall controllability and its elasticity, such that controllability was either low or elastic^{16–22}. The elasticity of control, however, must be dissociated from overall controllability to accurately diagnose mismanagement of resources. A given individual, for instance, may tend to overinvest resources because they overestimate controllability – for example, exercising due to a misguided belief that this can make one taller, when in fact height cannot be controlled. Alternatively, they may do so because they overestimate the elasticity of control – for example, a chess expert practicing unnecessarily hard to win against a novice, when their existing skill level already ensures control over the match's outcome. Identifying which misestimation drives resource mismanagement requires an experimental design that independently manipulates overall controllability from its elasticity.

Therefore, here we dissociate elastic and inelastic controllability using a novel treasure-hunt game where participants encountered three distinct environments: The first offered *high elastic controllability*, where a high level of control can only be attained by investing extra resources. The second offered *high inelastic controllability*, where control is attainable with any level of resource investment. And the third offered *low controllability*, where control was not attainable despite any level of resource investment. In each environment, participants were allowed to invest various amounts of resources to pursue reward or to altogether forgo pursuing reward.

Examining how participants adapted their resource investment to observed outcomes enabled us to determine whether and how participants inferred each environment's controllability and its elasticity.

Across two pre-registered studies (N = 514), we establish that people do indeed infer the elasticity of control. We present a computational model that explains how people make this inference, and identify individual biases in its implementation. We thus show that elasticity overestimation leads to an overinvestment of resources and is associated with elevated psychopathology involving an impaired sense of control. These findings establish elasticity as a crucial dimension of controllability that guides adaptive behavior, and a computational marker of control-related psychopathology.

Results

We had online participants (first study = 264, replication study = 250, ages =18-45, Mean 33 ± .6) play a novel treasure-hunt game set across multiple planets. On each trip (trial), participants attempted to travel from an initial location ('desert' or 'fountain') to a treasure (worth 150 coins; 15¢ of monetary compensation), located at either the house or the mountain (**Figure 2A**).

Participants could exercise control by boarding the train to reach the house, or the plane to reach the mountain (right-side **Figure 2B**). The present planet's controllability determined the probability that participants would succeed in boarding the vehicle they selected. In high-controllability planets, participants consistently boarded their preferred vehicle, enabling them to reliably reach their destination. Conversely, in low-controllability planets, participants frequently failed to board and were thus forced to walk to the nearest location. On 20% of trials, the nearest location was where the treasure was located, thus partially decoupling success in exerting control from reward.

To investigate how people learn the elasticity of control, we allowed participants to invest different amounts of resources in attempting to board their preferred vehicle. Participants could purchase one (40 coins), two (60 coins), or three tickets (80 coins) or otherwise walk for free to the nearest location. Participants were informed that a single ticket allowed them to board only if the vehicle stopped at the station, while additional tickets provided extra chances to board even after the vehicle had left the platform. For each additional ticket, the chosen vehicle appeared moving

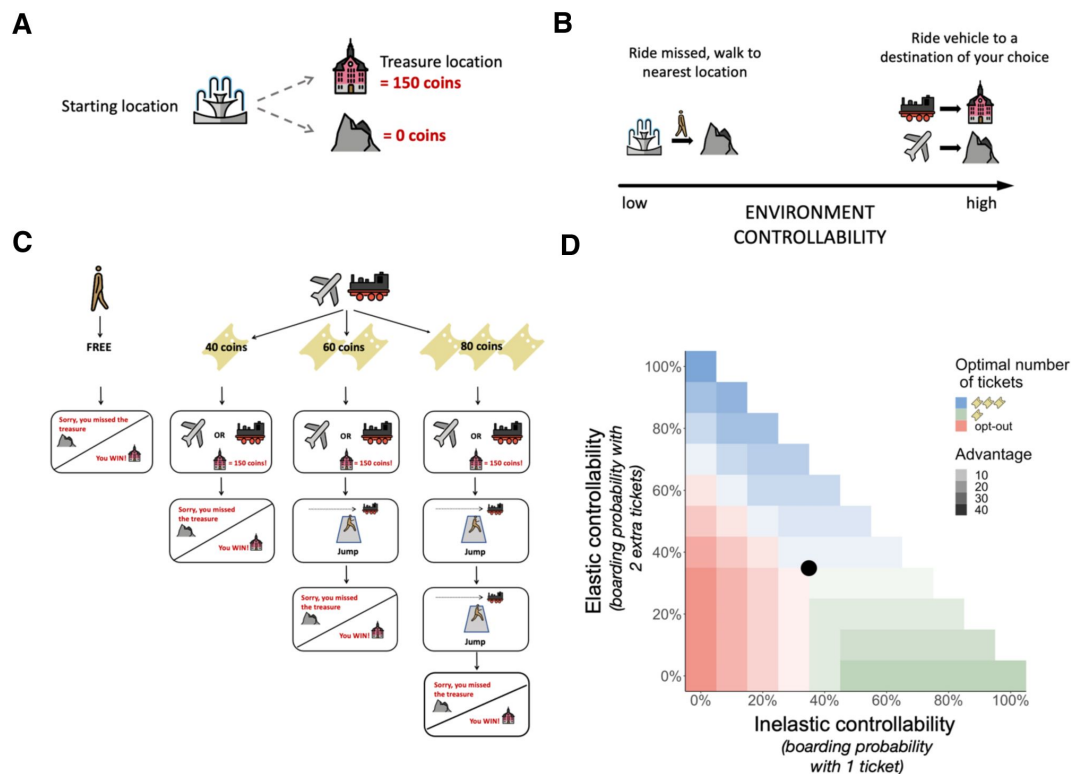


Figure 2.

Experimental design:

(A) **Goal.** On each trip to a planet, participants' goal was to reach a treasure located either at the house or the mountain (B) **Transition rules.** Participants could exercise control by boarding either the plane or the train to a destination of their choice, whereas missing the ride sent the participant walking to the nearest location (which happened to be the treasure location in only 20% of trials). (C) **Trip structure.** At the beginning of each trip, participants selected whether to purchase 1, 2, or 3 tickets to attempt to board their vehicle of choice, or walk for free to the nearest location. If the participant purchased at least one ticket, they were allowed to choose between the plane and the train. Then, for each additional ticket purchased, the participant was given an opportunity to increase their chances of boarding the vehicle. Specifically, the chosen vehicle appeared moving from left to right across the screen, and the participant attempted to board it by pressing the spacebar when it reached the center of the screen. At the end of the trip, participants were shown where they arrived at, allowing them to infer whether they successfully boarded the vehicle. (D) **The space of possible planets.** Planets varied in inelastic controllability – the probability that even one ticket would lead to successfully boarding the vehicle – and in elastic controllability – the degree to which two extra tickets would improve the probability of successfully boarding the vehicle (one extra ticket provided half the benefit). The color corresponds to the optimal number of tickets (0, 1 or 3) in each planet. The darker the color the higher the advantage in expected value gained by purchasing the optimal number of tickets relative to the second-best option. Each participant made 30 consecutive trips in one planet from the green area, one planet from the blue area, one planet from the red area, and one planet with identical characteristic across all participants (black circle).

from left to right across the screen, and participants could attempt to board it by pressing the spacebar when it reached the center of the screen. Thus, each additional ticket could increase the chance of boarding but also required a greater investment of resources—decreasing earnings, extending the trial duration, and demanding attentional effort to precisely time a button press when attempting to board.

Participants received no feedback about individual boarding attempts but were shown their initial and final locations at the end of each trip, enabling them to infer whether they had successfully boarded their vehicle. For simplicity, we defined inelastic controllability as the probability that even one ticket would lead to successfully boarding the vehicle, and elastic controllability as the degree to which two extra tickets would increase that probability.

To dissociate elastic from inelastic controllability, each participant visited three randomly sampled planets. In the first planet, controllability was high and inelastic, meaning that no matter how many tickets you purchased, you had a good chance of making your ride (sampled from green area in [Figure 2D](#), green). In the second planet, controllability was high and elastic, meaning that you needed to purchase two extra tickets to have a good chance of making your ride ([Figure 2D](#), blue). And in the third planet, controllability was low, meaning that no matter how many tickets you purchased, you did not have a good chance of making your ride ([Figure 2D](#), red).

Importantly, the order of these planets was randomized, and participants did not know in advance which planet they were visiting. Thus, they could only infer a planet's controllability and elasticity from the outcomes of their actions. Moreover, to increase sensitivity to individual biases, we had all participants visit a fourth planet where purchasing any number of tickets (0, 1, 2, 3) was equally optimal ([Figure 2D](#), black circle). We also homogenized participants' initial learning experiences by giving them three free tickets for the first five visits to every planet. As we shall see below, this latter measure also helped dissociate between different models of how participants solved the task.

People adapt to the elasticity of control

To determine whether participants inferred the elasticity of control, we first examined how they adapted their resource investment across planets. We found that participants were more likely to purchase tickets ('opt in') in more controllable planets, whether controllability was elastic (Initial and Replication studies, $p < .001$; mixed logistic regression) or inelastic (Initial and Replication studies, $p < .001$; [Figure 3A](#), green & C). Conversely, participants purchased more extra tickets only in planets with high elastic controllability (Initial $p = .01$, Replication $p = .004$; mixed probit regression; [Figure 3B](#), blue & C). Although substantial individual differences were observed in this regard ([Figure 3D](#)), these results show that most participants successfully adapted to both the controllability of the environments they visited and its elasticity.

People infer the elasticity of control

That participants adapted to each environment does not necessarily mean they inferred the elasticity of control (nor that they inferred control at all, but see Methods for how this was verified). Successful adaptation to the elasticity of control may also be achieved merely by treating each possible level of resource investment as a separate course of action. One could thus learn the degree of control afforded by each course of action (i.e., purchasing 1, 2, or 3 tickets), and then choose the one that affords the highest expected value considering its cost. This simple strategy could achieve optimal performance given unlimited experience. However, it would not be as efficient as a strategy that employs elasticity inference, because it does not make use of the dependencies that exist between different levels of resource investment in the degrees of control they afford. Thus, upon failing to board a vehicle despite purchasing 3 tickets, the simpler strategy would need to purchase 1 or 2 tickets to know these options would fail as well, whereas an agent

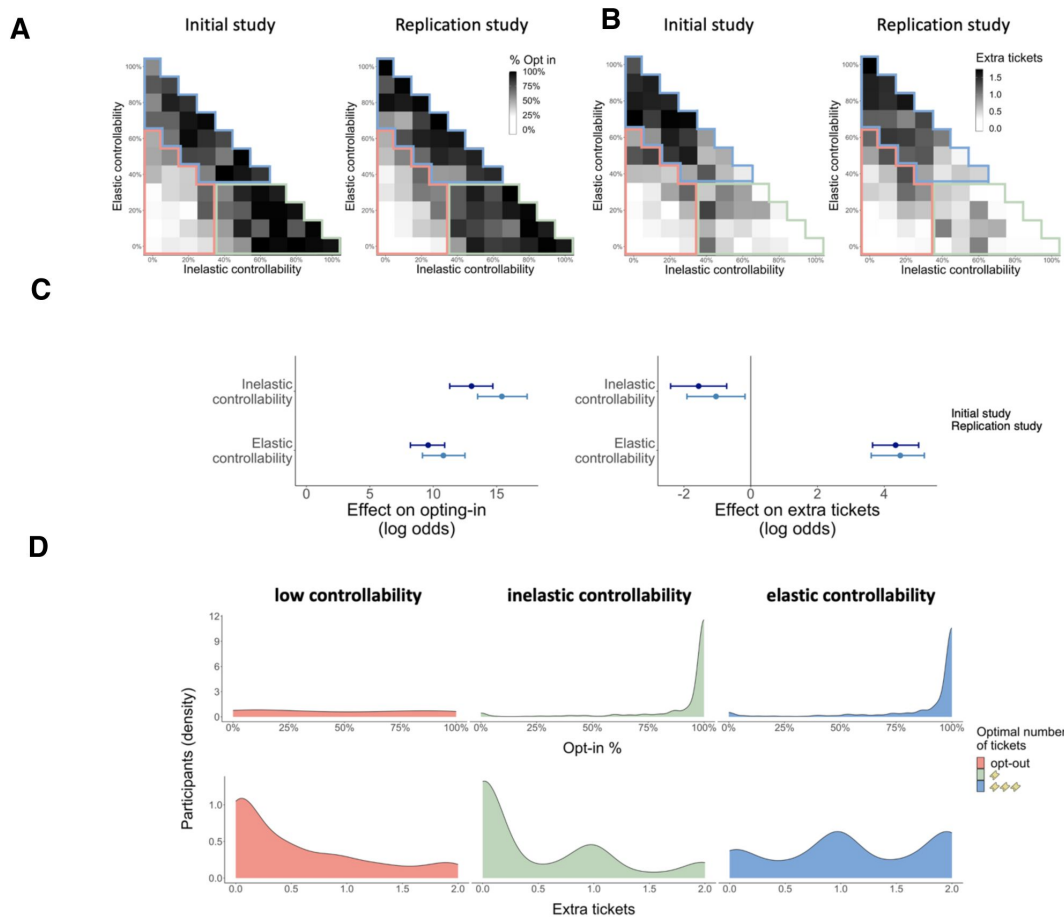


Figure 3.

Participants adapted to the elasticity of control:

Results from initial (N = 264) and replication (N = 250) studies. (A) **Opt-in percentage across all planets.** Participants opted-in more frequently on controllable planets (outlined in blue and green) than on uncontrollable planets (outlined in red) (B) **Extra tickets purchased across all planets.** Participants purchased more extra tickets in planets with high elastic controllability (outlined in blue). An average of 11 to 12 participants visited each planet in each of the studies. (C) **Effect of elastic and inelastic controllability on opting in and the purchasing of extra-tickets.** Bars show estimated fixed effects and 95% CIs from mixed logistic (opt-in) and probit (extra tickets) regressions. (D) **Individual differences.** Distribution of participants by opt-in rate (top) and average number of extra tickets purchased (bottom) across both studies, shown separately for planets with low controllability (red), high inelastic controllability (green), and high elastic controllability (blue).

equipped with the concept of elasticity would conclude that controllability is low, and it is thus best to opt out. Conversely, upon succeeding to board with 3 tickets, the simpler strategy would tend to favor purchasing 3 tickets, over 1 or 2 tickets, even if the extra tickets confer no benefit, whereas an agent that infers elasticity would not favor 3 over 1 or 2 tickets because it received no evidence that controllability is elastic (i.e., that the extra tickets help).

To determine which of these models best describes the way participants solved the task, we formalized the simpler strategy as the ‘controllability model’, and an elasticity-inferring strategy as the ‘elastic controllability model’ (pre-registered at: https://aspredicted.org/CHW_12H). Both models use beta distributions, characterized by parameters a and b , to represent the probability of boarding when purchasing different numbers of tickets, but only the elastic controllability model learns latent elasticity estimates. Specifically, in the ‘controllability model’, three beta distributions represent the expected level of control from purchasing 1, 2, or 3 tickets. Thus, the parameters of these distributions ($a_n, b_n, n \in \{1, 2, 3\}$) accumulate the number of times each number of tickets led (a_n) or did not lead (b_n) to successful boarding. Conversely, in the ‘elastic controllability model’, the beta distributions represent a belief about the maximum achievable level of control ($a_{\text{control}}, b_{\text{control}}$) coupled with two elasticity estimates that specify the degree to which successful boarding requires purchasing at least one ($a_{\text{elastic} \geq 1}, b_{\text{elastic} \geq 1}$) or specifically two ($a_{\text{elastic} = 2}, b_{\text{elastic} = 2}$) extra tickets. **Figure 4** shows schematically how the two models differ in updating their estimates in response to different types of outcomes. Both models similarly use these estimates to compute an expected value for purchasing 0, 1, 2, or 3 tickets, and accordingly choose between these options (see Methods for model equations and Supplementary Table 3 for all parameter fits, priors, and their interpretation).

Examining the predictions of both models, each simulated using the parameter setting that best fit participants’ choices, showed that the elastic-controllability model was better than the controllability model at opting-out in low controllability planets (elastic-controllability model % opt in = $49\% \pm .01$, controllability model % opt in = $63\% \pm .01$; **Figure 5C**, top panel). Most importantly, this higher level of performance closely matched participants’ behavior (% opt in = $47\%, \pm .02$). Furthermore, the elastic controllability model was also better at not purchasing extra tickets in low-controllability (elastic controllability extra tickets = $.49 \pm .01$ controllability extra tickets = $.65 \pm .01$) and inelastic-controllability (elastic controllability extra tickets = $.58 \pm .01$ controllability extra tickets = $.66 \pm .01$) planets, and this too was more closely aligned with participants’ behavior (low-controllability extra tickets = $51 \pm .02$, inelastic-controllability extra tickets = $53 \pm .02$; **Figure 5C**, bottom panel).

Beyond these general aspects of task behavior, having participants start on every planet with free 3-ticket trips also enabled us to examine their response to succeeding or failing with 3 tickets, prior to any other experiences on the planet (**Figure 5D**). As noted above, failure with 3 tickets makes the controllability model favor 1 and 2 tickets over 3 tickets, whereas success with 3 tickets makes it favor 3 tickets over 1 and 2 tickets. Critically, neither the participants nor the simulation of the elastic controllability model showed these effects. Upon failing with 3 tickets, both participants and the elastic model primarily favored to opt out. Conversely, if they succeeded with 3 tickets, they did not become less likely to purchase 1 or 2 tickets.

In fact, failure with 3 tickets even made participants favor 3, over 1 and 2, tickets. This favoring of 3 tickets continued until participants accumulated sufficient evidence about their limited control to opt out (Supplementary Figure 2). Presumably, the initial failures with 3 tickets resulted in an increased uncertainty about whether it is at all possible to control one’s destination. Consequently, participants who nevertheless opted in invested maximum resources to resolve this uncertainty before exploring whether control is elastic. While this behavior directly contradicts the controllability model, it also goes beyond the behavior exhibited by the elastic controllability model. To capture this aspect of participants’ behavior, we modified the elastic controllability

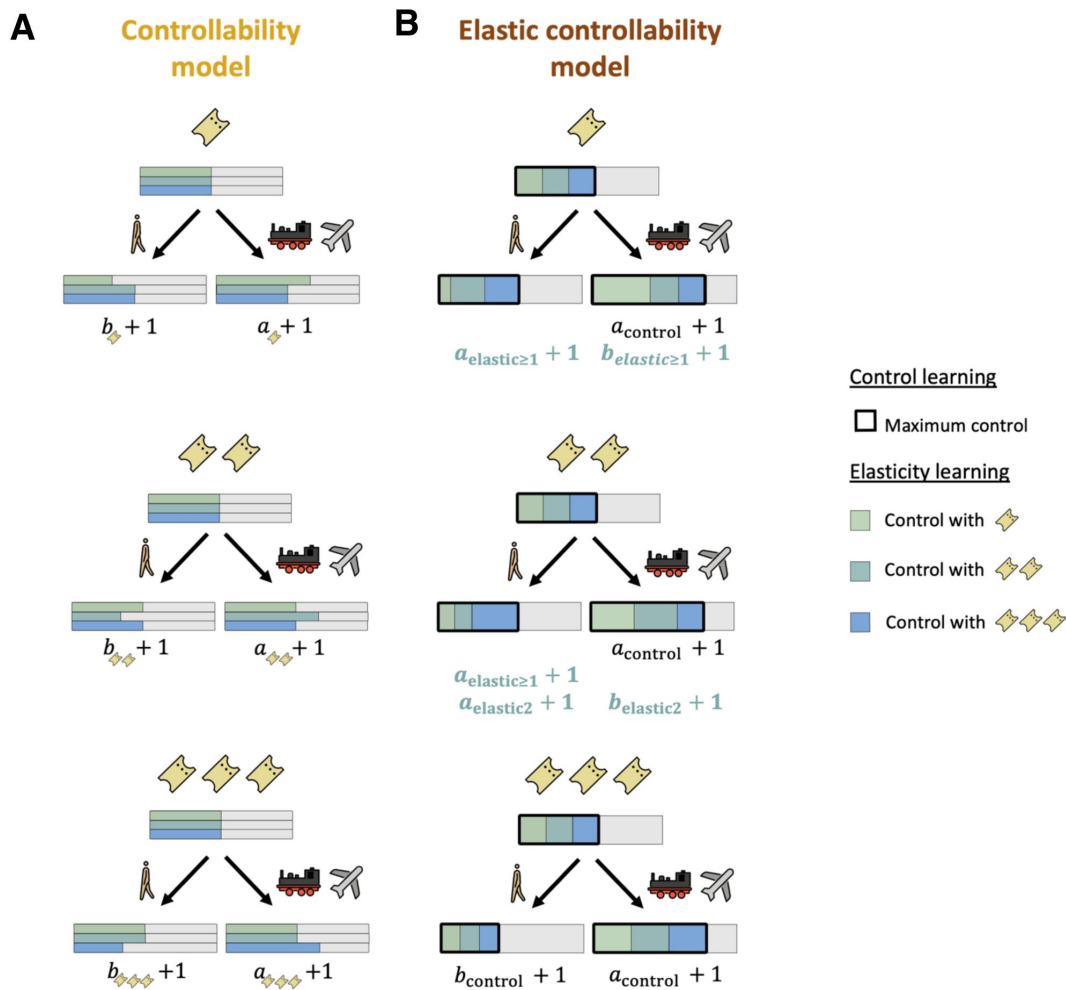


Figure 4.

Computational models:

Illustration of the learning rules of the *controllability* and *elastic controllability* models. The width of the colored region represents the estimated control, shown as a percentage of absolute control (gray area) along with the update rules based on each outcome. (A) **Controllability model:** This model treats the purchase of 1, 2, and 3 tickets as distinct actions. It accumulates evidence for the effectiveness (a) or lack of effectiveness (b) of each action based solely on whether or not the action led to successful boarding, illustrated by changes in the width of the shaded region corresponding to each ticket amount. (B) **Elastic Controllability model:** Represents beliefs about maximum controllability (black outline) and the degree to which at least one or two extra tickets are necessary to obtain it. These beliefs are used to calculate the expected control when purchasing 1 ticket (inelastic controllability) and the additional control afforded by 2 and 3 tickets (elastic controllability), demonstrated by the bars. The overlapping arrangement of the bars illustrates that, only in the elastic controllability model, maximum controllability is the sum of inelastic and elastic control. The model updates its beliefs as follows: **Top** – Successfully boarding with one ticket provides evidence of high maximum controllability ($a_{\text{control}} + 1$; expanded shaded region) and reduces the perceived need to purchase extra tickets ($b_{\text{elastic} \geq 1} + 1$; light green expanded). A failure to board does not change estimated maximum controllability, but rather suggests that 1 ticket might not suffice to obtain control ($a_{\text{elastic} \geq 1} + 1$; light green diminished). As a result, the model's estimate of control afforded by 1 ticket, as well as on average across ticket options, is reduced. **Middle** – Successfully boarding with 2 tickets provides evidence of high maximum controllability ($a_{\text{control}} + 1$; expanded shaded region), and reduces the perceived need to purchase two extra tickets ($b_{\text{elastic} \geq 2} + 1$; light & dark green expanded). Here too, a failure to board does not change estimated maximum controllability, but rather suggests that 2 tickets might not be sufficient, thus reducing the estimated control for 2 tickets and on average ($a_{\text{elastic} \geq 2} + 1$; light & dark green diminished). **Bottom** – Successfully boarding with 3 tickets provides evidence of high maximum controllability ($a_{\text{control}} + 1$; expanded shaded region), but offers no insight about whether fewer tickets would suffice to obtain it. Failure to board with 3 tickets provides evidence of low controllability ($b_{\text{control}} + 1$; diminished shaded region).

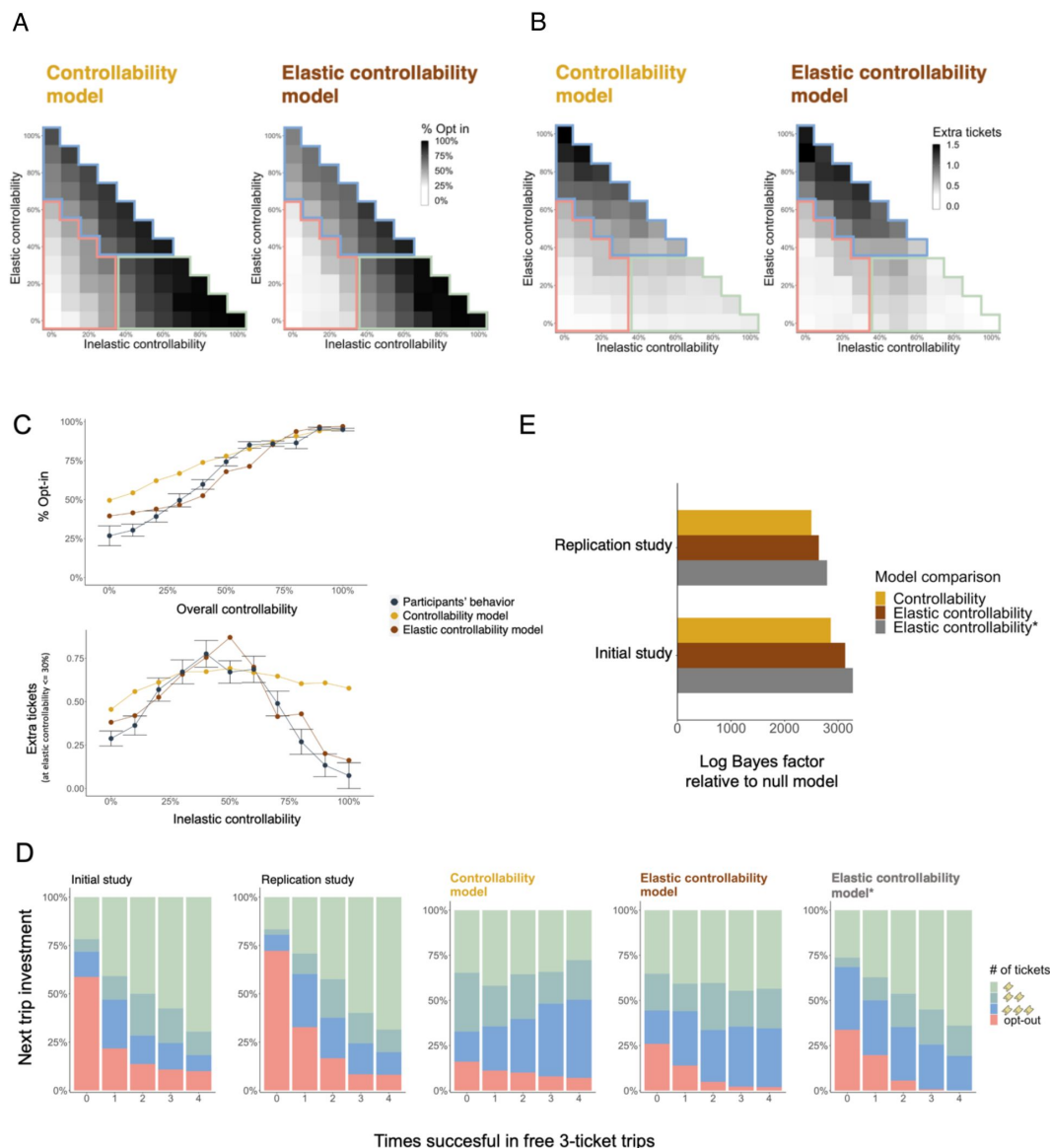


Figure 5.

Model simulations.

(A) **Opt-in.** Proportion of opt-in by the controllability and elastic controllability models across all planets. Each region is outlined by the color corresponding to the optimal ticket strategy (Blue: 3 tickets, Green: 1 ticket, Red: opt-out). (B) **Extra ticket purchases.** Extra tickets purchased by the controllability and elastic controllability models across all planets. Here too, each region is outlined by the color corresponding to the optimal ticket strategy. (C) **Comparison of opt-in rates and extra ticket purchases made by models and participants.** **Top** - Proportion of opting-in as a function of overall controllability. **Bottom** - The number of extra tickets purchased for each level of inelastic controllability when elasticity did not warrant purchasing extra tickets (i.e. below 30%). (D) **Resource investment following free 3-ticket trips.** The distribution of resource investment choices made immediately following the free 3-ticket trips, as a function of the outcomes of those trips. Results are shown for the two studies (Initial, Replication), the controllability model, and the original and modified variant of the elastic controllability models. (E) **Model comparison.** Consistency of each model with participant behavior is shown as log Bayes factor compared to a null model wherein participants have distinct preferences among different number of tickets to purchase but no latent controllability estimates.

model (elastic controllability*) such that it would increase its elasticity estimates when controllability is reduced. Model comparison consistently supported this modification (Initial study: log Bayes Factor = 143 Replication study: log Bayes factor = 155).

More importantly, both the original and modified elastic controllability models explained participants' behavior better than the controllability model (Initial: log Bayes Factor ≥ 270 , Replication: log Bayes Factor ≥ 148 ; **Figure 5E**) as well as other theoretically motivated alternatives (see Methods). These results establish that people infer the elasticity of control, and thereby adapt their resource allocation to the environment's controllability and its elasticity.

Individual elasticity biases explain misallocation of resources

To determine whether the elastic controllability* model could identify individual biases in elasticity inference, we fitted the model to participants' choices and thus derived each participant's beliefs about controllability ($\gamma_{\text{controllability}}$) and elasticity ($\gamma_{\text{elasticity}}$) prior to visiting any planet. $\gamma_{\text{controllability}}$ is used to initialize the model's belief about the maximum available control ($a_{\text{control}}, b_{\text{control}}$), with higher values reflecting stronger prior beliefs that planets are controllable. $\gamma_{\text{elasticity}}$ is used to initialize the model's elasticity estimates ($a_{\text{elastic}}, b_{\text{elastic}}$), with higher values reflecting stronger prior beliefs that control is elastic (see Methods). The stronger the belief the greater its influence on choices to opt in ($\gamma_{\text{controllability}}$) and purchase extra tickets ($\gamma_{\text{elasticity}}$) regardless of observed outcomes. In addition to these two model parameters, we examined four additional parameters that directly influence opting-in and the purchase of extra tickets. These include an inverse temperature parameter (β), which specifies the stochasticity with which expected values affect resource investment choices, and baseline- preference parameters for purchasing 1, 2, or 3 tickets ($\alpha_1, \alpha_2, \alpha_3$). Across both samples, shared variance among parameters never exceeded 14% (Supplementary Figure 1), and was particularly low for the controllability and elasticity priors (Initial study: 5%, Replication study: 3%). Thus, a tendency to assume the environment is controllable did not imply a tendency to assume that control is elastic, establishing inferences concerning maximum controllability and its elasticity as distinct facets of individual differences.

To clarify how fitted model parameters related to observable behavior, we regressed participants' opt-in rates and extra ticket purchases on the parameters (**Figure 6A**). We found that a higher controllability bias (higher $\gamma_{\text{controllability}}$) was associated with more opting in (Initial study $p < .001$, Replication study $p < .02$; logistic regression) and more extra tickets purchased (Initial $p < .001$, Replication $p < .001$; probit regression) on low-controllability, as compared to high- controllability, planets. Thus, a prior assumption that control is likely available was reflected in a futile investment of resources in uncontrollable environments.

In contrast, a higher elasticity bias was associated with the purchasing of more extra tickets on planets with inelastic, as compared to elastic, controllability (Initial study: $\beta = .18$, 95% CI = [0.03, 0.34], $p = .02$; Replication study: $\beta = .15$, 95%CI = [0.02 0.28], $p = .02$; linear regression). Thus, a prior assumption that control is likely elastic was primarily reflected in a needless investment of extra resources in environments where control could be obtained more cheaply.

Other parameters showed associations consistent with their roles in the model. Specifically, the inverse temperature parameter (β) was associated with a more optimal allocation of resources, all three baseline preference parameters ($\alpha_1, \alpha_2, \alpha_3$) were associated with opting in, and α_2 and α_3 , in particular, were also associated with the purchasing of more extra tickets (**Figure 6A**).

In sum, the model parameters captured meaningful individual differences in how participants allocated their resources across environments, with the controllability parameter primarily explaining variance in resource allocation in uncontrollable environments, and the elasticity parameter primarily explaining variance in resource allocation in environments where control was inelastic.

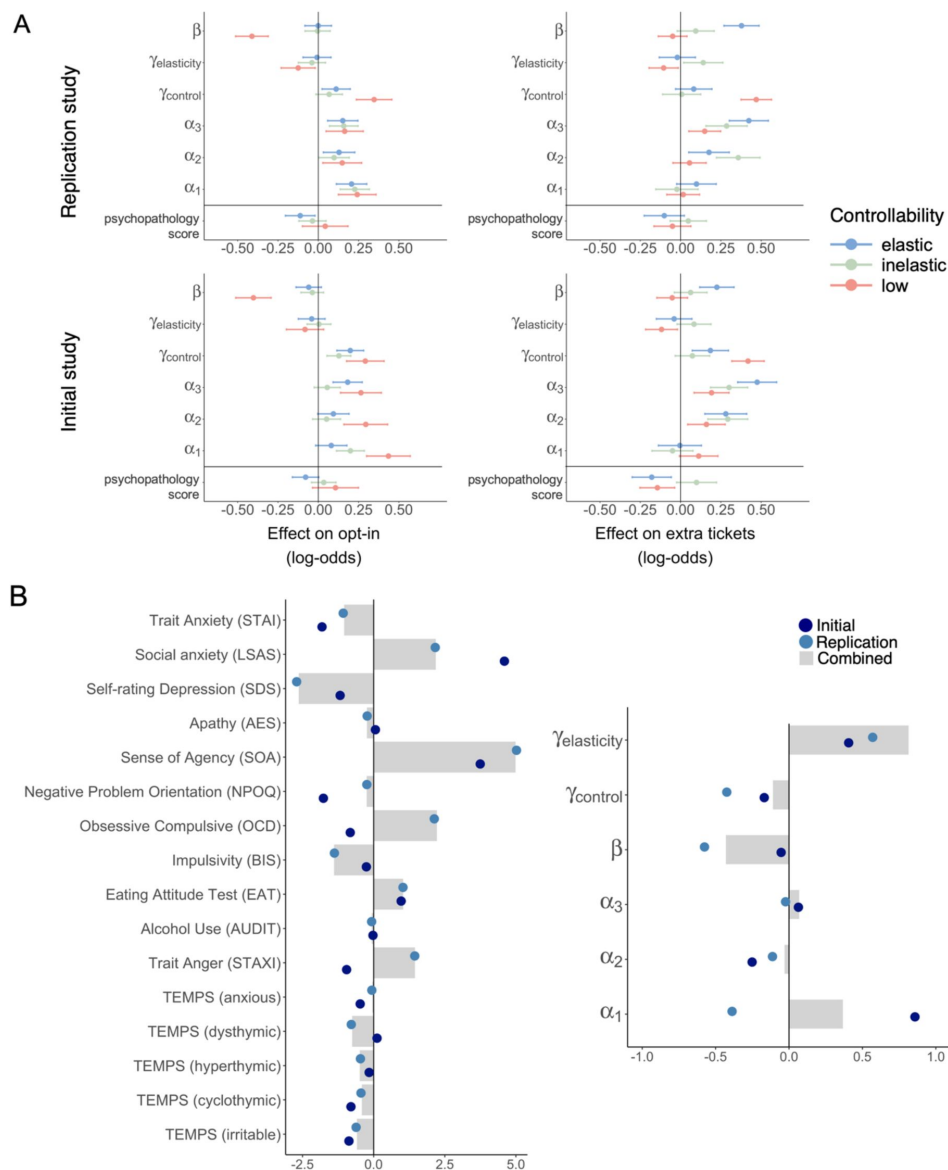


Figure 6.

(A) **Model parameters and task behavior.** Model parameters fitted to each participant's task behavior were used as predictors of opt-in (left; multiple logistic regressions) and extra ticket purchases (right; multiple probit regressions) separately for low controllability planets where opting out is optimal (red), high elastic controllability planets where purchasing 3 tickets is optimal (blue), and high inelastic controllability planets where purchasing 1 ticket is optimal (green). In a separate analysis, participants' composite psychopathology scores (see panel C) were similarly used to predict opt-in (left; logistic regression) and extra ticket purchases (right; probit regression). Bars represent regression coefficients and 95% CIs.

(B) **Model parameters and psychopathology.** Loadings from a Canonical Correlation Analysis (CCA) between model parameters and self-report psychopathology measures. Self-report measures were scored such that higher values reflect higher levels of psychopathology. Loadings are shown for the Initial (dark blue), Replication (light blue), and combined (gray bars) datasets. The magnitude of the loading reflects the degree to which each measure contributes to the correlation between parameters and self-reports.

(C) **Model parameters and psychopathology composite scores.** For each participant, a composite psychopathology score was calculated by multiplying their self-report scores with their CCA-derived loadings, and a composite model parameter score was calculated by multiplying their parameter fits and corresponding loadings. Each dot represents an individual participant. Black line: group-level posterior slope. Gray shading: standard error.

(D) **Significance testing.** p-values for the observed CCA correlation (circle) were computed against empirical null distributions (shaded area and bar showing 95% high-density interval). The top null distribution was obtained by shuffling the model parameter fits (top) relative to the psychopathology scores (top), the middle null distribution was obtained by shuffling only the elasticity prior parameter ($\gamma_{\text{elasticity}}$) while keeping all other parameters and self-report scores matched (middle), and the bottom null distribution was obtained by shuffling only the Sense of Agency score (SOA) while keeping all other parameters and self-report scores matched (bottom).

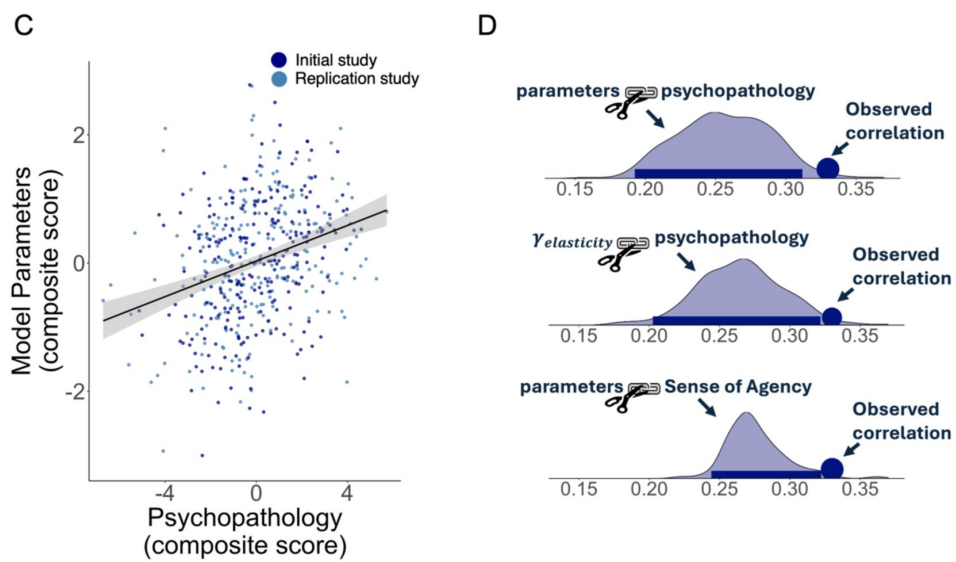


Figure 6. (continued)

Individual elasticity biases are associated with psychopathology

To examine whether the individual biases in controllability and elasticity inference have psychopathological ramifications, we assayed participants on a range of self-report measures of psychopathologies previously linked to a distorted sense of control (see Methods). Examining the direct correlations between model parameters and psychopathology measures (reported in Supplementary Figure 3) does not account for the substantial variance that is typically shared among different forms of psychopathology. For this reason, we instead used a canonical correlation analysis (CCA) to identify particular dimensions within the parameter and psychopathology spaces that most strongly correlate with one another. The CCA loadings indicate the contribution of each parameter of psychopathology score to the overall canonical correlation. We assessed statistical significance of the canonical correlation against a null distribution obtained by permuting the model parameter relative to the psychopathology scores. This showed that the correlation was significant in both the Initial ($p=.01$) and the Replication ($p=.02$) studies.

To obtain more robust estimates of the CCA loadings, we next ran a canonical correlation analysis on both datasets combined (Figure 6C, $\rho=.33$, $p=.004$). On the side of model parameters, the elasticity prior parameter ($\gamma_{\text{elasticity}}$) received the highest loading. By comparison, the controllability prior parameter received a lower loading with the opposite sign. A further permutation test showed that the contribution of the elasticity prior parameter to the parameter-psychopathology correlation was itself significant ($p=.04$; Figure 6D).

Loadings on the side of psychopathology were dominated by an impaired sense of agency (SOA; contribution to canonical correlation: $p=.03$, Figure 6D, bottom plot), along with obsessive compulsive symptoms (OCD), and social anxiety (LSAS) – all symptoms that have been linked to an impaired sense of control^{25–27}. Conversely, depression scores (SDS) received a negative loading, consistent with evidence of a lower willingness to expend one's resources in depression^{23,28–29}. This psychopathological profile also manifested in raw task measures, since, like the elasticity prior parameter, it too was associated with the purchasing of more tickets on planets where controllability was less elastic (Initial: $\beta = .19$, 95% CI = [.06, .32], $p=.004$; Replication: $\beta = .13$, 95% CI = [.01, .25], $p = .03$; linear regression; Figure 6A). Thus, a distinct psychopathological profile involving a distorted sense of agency was associated with a bias in elasticity inference.

Discussion

Across two pre-registered studies, we demonstrate that humans infer the elasticity of control over their present environment, and this informs their decisions of whether and how to act.

Accordingly, we found that a bias to overestimate the elasticity of control leads to a mismanagement of resources, and is associated with elevated psychopathology involving an impaired sense of agency.

Unlike prior work that treated controllability and its elasticity as coupled, our studies dissociated elastic and inelastic controllability, revealing that individuals who typically assume that taking action is worthwhile (overall controllability) do not necessarily assume that investing more resources in taking action is worthwhile (the elasticity of control). In fact, biases regarding controllability and its elasticity were oppositely related to psychopathology, revealing a psychopathological profile that combines underestimation of control with overestimation of the degree to which investing more resources is necessary to obtain control.

The observed elasticity biases across individuals are particularly striking given the simplicity of the experimental task in comparison to real-life scenarios. Unlike the task, where successful boarding directly determined a trip's outcome, in many real situations many actions may contribute to a single outcome. In this regard, the present work introduces a new kind of credit-assignment problem³⁰, which involves identifying not only which actions influenced the outcome but also which of these actions was made more effective by investing more resources in its execution. Moreover, whereas in the task elasticity increased linearly with the number of tickets, resource investment in real life often exhibits non-linear effects. Specifically, further increases in resource investment often result in diminishing returns, and beyond a certain point, even worse outcomes (e.g. hiring too many employees, making the work less efficient). Additionally, real life typically doesn't offer the streamlined recurrence of homogenized experiences that makes learning easier in experimental tasks, nor are people systematically instructed and trained about elastic and inelastic control in each environment. These complexities introduce substantial additional uncertainty into inferences of elasticity in naturalistic settings, thus allowing more room for prior biases to exert their influences. The elasticity biases observed in the present studies are therefore likely to be amplified in real-life behavior. Future research should examine how these complexities affect judgments about the elasticity of control to better understand how people allocate resources in real-life.

Our finding that elasticity biases are associated with agency-related psychopathologies raises interesting questions for future research on the causal nature of this association. One possibility is that biased elasticity inferences distort one's sense of agency by leading to maladaptive resource allocation that results in consistently suboptimal outcomes. Another possibility is that people who already have a reduced sense of agency, attempt to compensate for this feeling by investing more resources despite minimal benefit, thus demonstrating an elasticity bias.

Importantly, these possibilities are not mutually exclusive. That is, elasticity biases and a distorted sense of agency could reinforce each other in a cyclical manner. Longitudinal studies tracking or manipulating elasticity beliefs while monitoring changes in real-life resource investment and subjective sense of agency, could resolve how elasticity biases and psychopathology influence each other.

Whereas our pre-registered CCA effectively identified associations between task parameters and a psychopathological profile, this analysis method does not directly reveal relationships between individual variables. Auxiliary analyses confirmed significant contributions of both elasticity bias and sense of agency to the observed canonical correlation, but the contribution of other measures remains to be determined by future work. Such work could employ other established measures of agency, including both behavioral indices and subjective self-reports, to better understand how these constructs relate across different contexts and populations.

Our operationalization of controllability aligns with previous work that defined controllability as the degree to which actions influence the probability of obtaining a reward^{5-6,6}. Other notable works defined controllability as the degree to which actions predict subsequent states, independent of reward¹⁻³. Here we adhered to the former definition due to our present focus on resource elasticity, since individuals are unlikely to invest resources to gain control if control does not yield more reward. That said, we followed the latter work in ensuring that controllability is not confounded with predictability², in the sense that in planets with high controllability, outcomes were not necessarily more predictable than in planets with low controllability (See Methods). This enabled us to specifically assess individual inferences concerning controllability and its resource elasticity.

Future research could explore alternative models for implementing elasticity inference that extend beyond our current paradigm. First, further investigation is warranted concerning how uncertainty about controllability and its elasticity interact. In the present study, we minimized

individual differences in the estimation of maximum available control by providing participants with three free tickets at their initial visits to each planet. We made this design choice to isolate differences in the estimation of elasticity, as opposed to maximum controllability. To study how these two types of estimations interact, future work could benefit from modifying this aspect of our experimental design.

Second, future models could enable generalization to levels of resource investment not previously experienced. For example, controllability and its elasticity could be jointly estimated via function approximation that considers control as a function of invested resources. Although our implementation of this model did not fit participants' choices well (see Methods), other modeling assumptions or experimental designs may offer a better test of this idea.

Another interesting possibility is that individual elasticity biases vary across different resource types (e.g., money, time, effort). For instance, a given individual may assume that controllability tends to be highly elastic to money but inelastic to effort. Although the task incorporated multiple resource types (money, time, and attentional effort), the results may differ depending on the type of resources on which the participant focuses. Future studies could explore this possibility by developing tasks that separately manipulate elasticity with respect to different resource types. This would clarify whether elasticity biases are domain-specific or domain-general, and thus elucidate their impact on everyday decision-making.

Methods

Pre-registration

Both the Initial and Replication studies were pre-registered prior to data collection. The pre-registrations included hypotheses, sample sizes, exclusion criteria, task design, analysis plans (regressions, CCA), and computational models. Detailed pre-registration documents are available at https://aspredicted.org/Q7M_BHR and https://aspredicted.org/CHW_12H.

Participants

We used the Prolific online platform (prolific.com) to recruit participants (Initial study $N = 264$, Replication study $N = 250$). Participants were selected based on their demonstrated high engagement levels in prior studies, fluency in English, and absence of psychiatric diagnoses. Due to cultural differences in the perception of controllability³¹, participants were recruited from English speaking western countries. The average age of participants across both datasets was 33 years old ($SE = .6$), and 38% were female.

The sample size was determined using a power analysis as described below. Exclusion criteria were applied to ensure data quality and participant engagement. Specifically, participants were excluded if they demonstrated inattentiveness (defined as more than 8 mistakes on comprehension quizzes during the instruction phase), displayed less than 90% accuracy in selecting the correct vehicle corresponding to its pre-learned destination (mean error rate = 3%, range [0-10%]; indicating a lack of task comprehension), or chose not to purchase any tickets in 90% or more of trials across all experimental conditions (suggesting a lack of engagement with the task). These criteria led to the exclusion of 22 participants (8.3% exclusion rate) from the initial study and 23 participants (8.9% exclusion rate) from the replication study. Participants gave written informed consent before taking part of the study, which was approved by the university's ethics review board (approval number: 2022-06213). Participants were paid at a rate of 9 euros per hour.

Experimental Task

Participants were informed about the location of a treasure (150 coins, worth 15¢ of bonus monetary compensation) and could exercise control by boarding the train to reach the house, or the plane to reach the mountain. (see main text and [Figure 2](#)). Forgoing or failing to board resulted in walking to the nearest location which happened to be where the treasure was located 20% of the time.

The first ticket (cost = 40 coins) allowed participants to select their desired vehicle, but it only stopped for them at the platform some percentage of the time, corresponding to the inelastic controllability of the planet. Participants could purchase up to 2 additional tickets (cost = 20 coins each) to attempt boarding the moving vehicle after it had left the platform. This improved the probability of boarding in correspondence with the elastic controllability of the planet.

When participants purchased a second or third ticket, the chosen vehicle appeared moving from left to right across the screen, and participants attempted to board it by pressing the spacebar when it reached the center of the screen. The precision of each boarding attempt was quantified as the absolute distance of the vehicle from the screen's center at the time of the press. Unless elastic control was below 15%, boarding attempts with precision in the top 15 percentile relative to the participant's prior performance automatically succeeded, and unless elastic control was above 85%, those in the bottom 15 percentile automatically failed. For the remaining attempts, the outcome was probabilistically determined such that the overall probability of successfully boarding matched the elasticity of control on that planet. To verify that participants' pressing ability did not bias their preference for purchasing extra tickets, we ran a mixed probit regression of the number of extra tickets purchased on participants' average press precision. We found no significant effect ($p = .45$). When participants purchased multiple tickets, they made all boarding attempts in sequence without intermediate feedback, only learning whether they successfully boarded upon reaching their final destination. This served two purposes. First, to ensure that participants would consistently need to invest not only more money but also more effort and time in planets with high elastic controllability. Second, to ensure that results could potentially generalize to the many real-world situations where the amount of invested effort has to be determined prior to seeing any outcome (e.g., preparing for an exam or a job interview).

At the end of each trip, participants were shown where they arrived, allowing them to infer whether they successfully boarded their vehicle. Since this feedback is only interpretable if participants knew each vehicle's destination, we only included participants in subsequent analyses if they chose the vehicle corresponding to the destination of the treasure with at least 90% accuracy (Mean 97% $\pm$ 2%, see first exclusion criteria). The average trial duration was approximately 11 seconds (SE = 0.05), with each additional ticket extending this duration by approximately 0.5 seconds (SE = 0.04), increasing the time investment required from participants.

To allow sufficient learning of each planet's elasticity and controllability, participants took 30 consecutive trips on each planet. This trip count was determined based on pilot data (N=19) by analyzing when participants' strategies stabilized. Specifically, we examined the standard deviation in ticket choices over the final 5 trips and found stabilization after around 15 trips (STD = 0.33 compared to .63 in the first 15 trips). Based on this, we provided 15 initial trips per planet for learning, and analyzed ticket choices starting from trip 16 onward. This approach followed our pre-registered plan.

To homogenize learning experience across participants, we gave participants three free tickets for the first five visits to every planet. However, only four of those trips were informative, since on one of every five trips (randomly placed), the treasure was located in the nearby location, making boarding inconsequential for reaching it.

Points accumulated across all planets throughout the session, with participants explicitly motivated to maximize their total points as this directly determined their monetary bonus payment. To ensure that accumulated gains did not lead participants to adopt a simple heuristic strategy of always purchasing 3 tickets, we conducted a mixed probit regression examining whether the number of accumulated coins influenced participants' decisions to purchase extra tickets. We did not find such an effect ($\beta_{\text{coin accumulated}} = .01$ $p = .87$), ruling out the potential strategy shift.

Planet types

To study elasticity learning across all possible degrees of controllability and elasticity, 66 unique planet conditions were generated by systematically varying inelastic controllability (the probability of the chosen vehicle stopping at the platform if purchasing a single ticket) and elasticity (the added probability of boarding from purchasing 2 extra tickets) from 0 to 100% in 10% increments. The treasure value (150 coins) and ticket costs (40 coins for the first ticket, 20 coins for extra tickets) were chosen so that the optimal strategy, as determined by expected value calculations ([equation 1](#)), was to purchase 0 tickets on 22 planets (low controllability condition), 1 ticket on a different set of 22 planets (high inelastic controllability condition), and 3 tickets on the remaining 22 planets (high elastic controllability condition). Participants were randomly assigned one planet from each condition, while ensuring they could earn roughly the same total number of coins if they made optimal choices (Initial study = $45 \pm .3$, Replication study $46 \pm .3$). To address potential confounds between controllability and predictability², we designed the experiment such that high-controllability planets (where purchasing 1 or 3 tickets was optimal) were not inherently more predictable than low-controllability planets. We quantified predictability as the entropy of the probability distribution determining whether the participant will or will not arrive at the rewarded location. For 60% of participants, purchasing tickets on high-controllability planets decreased entropy (i.e., increased predictability) compared to low-controllability planets, whereas for the other 40% it increased entropy (i.e., decreased predictability). To increase sensitivity to individual differences, all participants first encountered a planet where purchasing any number of tickets (0, 1, 2, or 3) yielded the same expected value (EV = 30 coins).

Initial training

Prior to engaging in the task, participants completed 90 practice trials. These practice trials were divided equally across planets with high inelastic controllability, high elastic controllability, and low controllability. To reinforce participants' understanding of how elasticity and controllability were manifested in each planet, they were informed of the planet type they had visited after every 15 trips. Specifically, for high inelastic controllability planets, the message read: “*Your ride actually would stop for you with 1 ticket. So purchasing extra tickets, since they do cost money, is a WASTE.*” For low controllability planets: “*Doesn't seem like the vehicle stops for you nor does purchasing extra tickets help.*” Lastly, for high elastic controllability planets: “*Hopefully by now it's clear that only by purchasing 3 tickets (LOADING AREA) are you consistently successful in catching your ride.*”.

Power analysis

The sample sizes for both the initial and the replication studies were pre-registered before data collection commenced. For the initial study, we derived the sample size of 264 participants using a power analysis aiming to estimate with high accuracy (credible interval width < 0.1) the likelihood that participants will opt-in for any combination of elastic and inelastic control (66 total combinations). For this purpose, we used Kruschke's (2014) procedure for Bayesian Power Analysis³². Based on pilot data (N=19), we fitted a Bayesian Beta Binomial model to the proportion of opt-ins in the planet with no optimal strategy (and therefore the highest variability in behavior would be expected), and found that 12 participants were required for the credible

interval of the binomial proportion parameter to be lower than 0.1. Since each participant completes 3 combinations of elastic and inelastic control, we multiplied the required sample size by the total number of planets in each condition (22) which results in an estimate of $n=264$.

To examine whether this sample size is sufficient to detect significant effects of elastic and inelastic controllability on opting in (**Figure 3A,C**), and the purchasing of extra tickets (**Figure 3B,C**), we used a bootstrap procedure whereby we sampled 264 participants from the pilot data and ran the regression analyses on them. We repeated this procedure for 50 iterations, each time recording β_{elastic} , $\beta_{\text{inelastic}}$, and their p-values. For analysis 1, we found that both β_{elastic} and $\beta_{\text{inelastic}}$ were significantly above 0 in all iterations, and for analysis 2, β_{elastic} was significantly above 0 in all iterations, thereby giving us >98% power to detect all of our effects of interest. Confidence intervals were derived from a binomial test in which a positive significant coefficient counted as a success.

For the replication study, we derived the sample size of 250 participants using a power analysis aiming for 95% power to detect a significant ($p < .05$) canonical correlation between the model-derived parameters and questionnaire scores (**Figure 6B**). For this purpose, we sampled with replacement from the initial dataset ($n=264$) which had showed a significant canonical correlation ($p=.01$; **Figure 6B**), and calculated the p-value of a CCA in the subsample. We repeated this procedure for 100 iterations, each time recording the calculated p-value. To ensure our sample size was sufficient to detect significant effects in the regression models predicting opt-in and extra ticket purchases from the composite psychopathology score with over 95% power (**Figure 6C**) we conducted another bootstrap analysis. We sampled 250 participants with replacement each time calculating the composite score and ran the regression analyses on that score, repeating this procedure 100 times. We found that $\beta_{\text{composite}}$ was significant ($p < .05$) in 95% of iterations (95% power).

Regression analyses

Mixed model regression analyses were performed using the “ordinal” package which fits cumulative mixed models via Laplace approximations³³. The model predicts the likelihood of purchasing tickets (logistic model) and how many extra tickets are purchased (probit model) on the last 15 trips on each planet. Predictors included fixed and random effects for elastic and inelastic controllability and fixed and random intercepts per participant and planet number, with the latter included to account for potential learning or fatigue effects. Statistical significance was assessed using the Wald Z Test³⁴ with normality assessed using the Shapiro-Wilk Test³⁵.

Computational Modeling

To assess whether participants inferred elasticity or merely learned the control afforded by each course of action separately, we fit two competing reinforcement-learning models (‘controllability’ and ‘elastic controllability’). Both models use beta distributions defined by parameters (a , b), to form controllability beliefs for each ticket quantity n ($n \in \{1, 2, 3\}$), but only the elastic controllability model leverages the dependencies that exist between different levels of resource investment. Below, we expand on the computations underlying both models.

Controllability model

The controllability model accumulates the number of times that each ticket quantity n led to successful or unsuccessful boarding:

$$\begin{aligned} a_n &\leftarrow a_n + 0 \\ b_n &\leftarrow b_n + (1 - 0) \end{aligned} \quad (\text{equation 1})$$

Where $O=1$ if boarding was successful and $O=0$ if not. The probability of boarding with 1,2 or 3 tickets on planet s is thus simply the expected value of that particular beta distribution:

$$p(\text{boarding}|n, s) = \frac{a_n}{a_n + b_n}. \quad (\text{equation 2})$$

Here and in all subsequent models, learning only takes place when the treasure is not located on the nearby planet, which makes it possible to identify boarding success.

Elastic controllability model

The *elastic controllability model* explains ticket choices based on latent beliefs about the controllability and elasticity of each planet (see **Figure 4**). These beliefs are represented by three beta distributions, each defined by two parameters (a , b). The expected value of one beta distribution (defined by a_{control} , b_{control}) represents the belief that boarding is possible (controllability) with any number of tickets whereas the expected value of the other two beta distributions represent the belief that successful boarding is added by purchasing at least one ($a_{\text{elastic} \geq 1}$, $b_{\text{elastic} \geq 1}$) or specifically two ($a_{\text{elastic} \geq 2}$, $b_{\text{elastic} \geq 2}$), extra tickets (elasticity).

Thus, the parameter a_{control} accumulates the number of successful boarding attempts with any number of tickets, whereas b_{control} accumulates the number of failed boarding attempts despite purchasing the maximum number of tickets (3 tickets):

$$\begin{aligned} a_{\text{control}} &\leftarrow a_{\text{control}} + 1 \text{ if win with 1,2, or 3 tickets} \\ b_{\text{control}} &\leftarrow b_{\text{control}} + 1 \text{ if lose with 3 tickets} \end{aligned} \quad (\text{equation 3})$$

A failure to board with fewer tickets doesn't reduce the maximum controllability estimate but instead suggests that more tickets or the maximum number of tickets may be needed to obtain control:

$$\begin{aligned} a_{\text{elastic} \geq 1} &\leftarrow a_{\text{elastic} \geq 1} + 1 \text{ if lose with 1 or 2 tickets} \\ a_{\text{elastic} \geq 2} &\leftarrow a_{\text{elastic} \geq 2} + 1 \text{ if lose with 2 tickets} \end{aligned} \quad (\text{equation 4})$$

Conversely, successfully boarding with fewer tickets is evidence that obtaining controllability does not require maximal or any extra tickets:

$$\begin{aligned} b_{\text{elastic} \geq 1} &\leftarrow b_{\text{elastic} \geq 1} + 1 \text{ if win with 1 ticket} \\ b_{\text{elastic} \geq 2} &\leftarrow b_{\text{elastic} \geq 2} + 1 \text{ if win with 2 tickets} \end{aligned} \quad (\text{equation 5})$$

To account for our finding that failure with 3 tickets made participants favor 3, over 1 and 2, tickets, we introduced a modified elastic controllability* model, wherein purchasing extra tickets is also favored upon receiving evidence of low controllability (loss with 3 tickets). This effect was modulated by a free parameter κ which reflects a tendency to prioritize resolving uncertainty about whether control is at all possible by investing maximum resources:

$$\begin{aligned} a_{\text{more}} &\leftarrow a_{\text{elastic} \geq 1} + \kappa \text{ if lose with 3 tickets} \\ a_{\text{elastic} \geq 2} &\leftarrow a_{\text{elastic} \geq 2} + \kappa \text{ if lose with 3 tickets} \end{aligned} \quad (\text{equation 6})$$

The estimated controllability and elasticity, as characterized by the expected value of the three beta distributions ($C_{\text{total control}}$, C_{more} , C_{full}), is used to derive the probability of boarding the selected vehicle with 1, 2, or 3 tickets. Thus, the estimated probability of boarding with 1 ticket is an estimate of inelastic control, that is, the percentage of the total control $C_{\text{total control}}$ that does not require investing more resources:

$$P(\text{boarding}|n = 1) = C_{\text{total control}} \times (1 - C_{\text{more}}) \quad (\text{equation 7})$$

For 2 tickets, the added probability of boarding is an estimate of what percentage of total control is elastic (C_{more}) but does not require investing maximal resources ($1 - C_{\text{full}}$):

$$P(\text{boarding}|n = 2) = P_s(\text{boarding}|n = 1) + C_{\text{total control}} \times C_{\text{more}} \times (1 - C_{\text{full}}) \quad (\text{equation 8})$$

Finally, since 3 tickets is the maximum resource investment, the estimated probability of boarding with 3 tickets is equivalent to an estimate of total attainable control.

$$P(\text{boarding}|n = 3, s) = C_{\text{total control}} \quad (\text{equation 9})$$

Models' resource investment

Because participants win 20% of the time even when they fail to board their selected ride, we calculate the probability of arriving at the treasure with n tickets as:

$$P(\text{win}|n) = P(\text{boarding}|n) + .2 \times (1 - P(\text{boarding}|n)) \quad (\text{equation 10})$$

Then, taking the possible reward ($R = 150$) and ticket costs ($C = \{40, 20, 20\}$) into account, the expected value for purchasing n tickets ($n = \{1, 2, 3\}$) tickets is:

$$V(n) = (R - \sum_{i=1}^n C_i) \times P(\text{win}|n) - \sum_{i=1}^n C_i \times (1 - P(\text{win}|n)) \quad (\text{equation 11})$$

One exception to this rule is the expected value of opting out, $V(0)$, which is similar across all planets, and equals the potential reward (150) times the probability that walking will lead to the reward (0.2).

The probability that a participant will purchase 0, 1, 2, or 3 tickets is calculated using the SoftMax function such that:

$$p(n) \propto \beta \times V(n) + \alpha_n + \rho \times I(n_{t-1} == n) \quad (\text{equation 12})$$

where β is an inverse temperature parameter, α_{1-3} are parameters that represent participants' fixed tendencies to purchase 1, 2, or 3 tickets respectively, and ρ is a perseveration parameter that represents participant tendency to purchase the same number of tickets as in the last trial (n_{t-1}). A_0 is set to 0 to maintain parameter identifiability.

We compared the aforementioned models to a null model wherein participants have distinct preferences among different number of tickets (α_{1-3}) and perseveration (ρ) but no latent controllability estimates ($\beta = 0$ in [equation 1](#) [2](#)). We also evaluated variants of both models that replaced the SoftMax-based exploration (governed by inverse temperature β) with a decaying epsilon-greedy approach. This epsilon-greedy variant, which begins with a fixed exploration rate that decreases with the number of trials⁵⁹, performed significantly worse at explaining participants' choices compared to the SoftMax-based models.

Prior controllability and elasticity beliefs

To capture prior biases that planets are controllable and elastic, we introduced parameters $\gamma_{\text{controllability}}$ and $\gamma_{\text{elasticity}}$, each computed by multiplying the direction ($\lambda - 0.5$) and strength (ϵ) of individuals' prior belief. $\lambda_{\text{controllability}}$ and $\lambda_{\text{elasticity}}$ range between 0 and 1, with values above 0.5 indicating a bias towards high controllability or elasticity, and values below 0.5 indicating a bias towards low controllability or elasticity. $\epsilon_{\text{controllability}}$ and $\epsilon_{\text{elasticity}}$ are positively valued parameters capturing confidence in the bias. Parameter recovery analyses confirmed both good recoverability (see S2 Table) and low confusion between bias direction and strength ($\epsilon_{\text{controllability}} \rightarrow \lambda_{\text{controllability}} = -.07$, $\lambda_{\text{controllability}} \rightarrow \epsilon_{\text{controllability}} = -.16$, $\epsilon_{\text{elasticity}} \rightarrow \lambda_{\text{elasticity}} = .15$,

$\lambda_{\text{elasticity}} \rightarrow \epsilon_{\text{elasticity}} = .04$), ensuring that observed biases and their relation to psychopathology do not merely reflect slower learning (Supplementary Figure 4), which can result from changes in bias strength but not direction.

For each new planet participants encountered, these parameters were used to initialize the beta distributions representing participants' beliefs:

$$\begin{aligned} a_{\text{controllability}} &= 2 \times \epsilon_{\text{controllability}} \times \lambda_{\text{controllability}} \\ b_{\text{controllability}} &= \epsilon_{\text{controllability}} \times (1 - \lambda_{\text{controllability}}) \\ a_{\text{more,full}} &= \epsilon_{\text{elasticity}} \times \lambda_{\text{elasticity}} \\ b_{\text{more,full}} &= \epsilon_{\text{elasticity}} \times (1 - \lambda_{\text{elasticity}}) \end{aligned} \quad (\text{equation 13})$$

$a_{\text{controllability}}$ is multiplied by 2 so that in the absence of a bias, it would have equal preference for purchasing 0 to 3 tickets (this requires that $C_{\text{controllability}} = \frac{2}{3}$, $C_{\text{more,full}} = \frac{1}{2}$).

For consistency, in the 'controllability model', the controllability estimates for each ticket quantity are initialized for each ticket quantity n , where $n=\{1,2,3\}$, as:

$$\begin{aligned} a_n &= \epsilon_n \times \lambda_n \\ b_n &= \epsilon_n \times (1 - \lambda_n) \end{aligned} \quad (\text{equation 14})$$

Here, to ensure no preference between different ticket quantities in the absence of a bias, α_1 is multiplied by 2 and α_2 is multiplied by $\frac{3}{2}$.

Testing alternative models

To further validate our elastic controllability model, we tested it against several theoretically motivated alternatives that could potentially explain participants' behavior.

To ascertain that participants were truly learning latent estimates of controllability rather than simpler associations, we conducted two complementary analyses.

First, we implemented a simple Q-learning model that directly maps ticket quantities to expected values based on reward prediction errors, without representing latent controllability. This associative model performed substantially worse than even our simple controllability model (log Bayes Factor ≥ 1854 on the combined datasets). Second, we fitted a variant of the elastic controllability model that compared learning from control-related versus chance outcomes via separate parameters (instead of assuming no learning from chance outcomes). Chance outcomes were observed by participants in the 20% of trials where reward and control were decoupled, in the sense that participants reached the treasure regardless of whether they boarded their vehicle of choice. Results showed that participants learned considerably more from control-related, as compared to chance, outcomes (mean learning ratio=1.90, CI= [1.83, 1.97]). Together, these analyses show that participants were forming latent controllability estimates rather than direct action-outcome associations.

Having established that participants indeed learn latent controllability estimates, we next examined our model's interpretation of the κ parameter. Our model accounts for the observation that failures with 3 tickets made participants more likely to purchase 3 tickets when they opted in. To test whether this behavior resulted from participants not accepting that the planet is uncontrollable, we implemented a variant of the model where κ not only increases the elasticity estimate but also reduces the controllability update (using $\beta_{\text{control}} + (1 - \kappa)$ instead of $\beta_{\text{control}} + 1$) after failures with 3 tickets. However, implementing this coupling diminished the model's fit to the data (log Bayes Factor ≥ 48), as compared to allowing both effects to occur independently, indicating that the increase in 3 ticket purchases upon failing with 3 tickets did not result from participants

not accepting that controllability is in fact low. Thus, we maintained our original interpretation that failure with 3 tickets increases uncertainty about whether control is possible at all, leading participants who continue to opt in to invest maximum resources to resolve this uncertainty.

Having validated our interpretation of κ , we next examined whether participant behavior would be better characterized as a continuous function approximation rather than the discrete inferences in our model. To test this, we implemented a Bayesian model where participants continuously estimate both controllability and its elasticity as a mixture of three archetypal functions mapping ticket quantities to success probabilities. The flat function provides no control regardless of how many tickets are purchased (corresponding to low controllability). The step function provides full control as long as at least one ticket is purchased (inelastic controllability). The linear function linearly increases control with the number of extra tickets (i.e., 0%, 50%, and 100% control for 1, 2, and 3 tickets, respectively; elastic controllability). The model computes the likelihood that each of the functions produced each new observation, and accordingly updates its beliefs. Using these beliefs, the model estimates the probability of success for purchasing each number of tickets, allowing participants to weigh expected control against increasing ticket costs. Despite its theoretical advantages for generalization to different ticket quantities, this continuous function approximation model performed significantly worse than the elastic controllability model (log Bayes Factor > 4100 on combined datasets), suggesting that participants did not assume that control increases linearly with resource investment.

Model fitting

Reinforcement-learning models were fitted to all choices made by participants via an expectation maximization approach used in previous work^{36–38}. We generated random parameter settings drawn from predefined group-level distributions and calculated the likelihood of observing the participants' choices given each parameter setting. We then approximated the posterior estimates of the group-level prior distributions for each parameter by resampling the parameter values, weighted by their respective likelihoods, and refitted the data based on the updated priors. This iterative process of resampling and refitting continued until no further improvement to model fit. To determine the best-fitting parameters for each individual participant, we computed a weighted average of the final batch of parameter settings, where each setting was weighted by the likelihood it assigned to the individual participant's choices.

Model parameter β was initialized by sampling from a Gamma distribution ($k = 1$, $\theta = 1$), λ was initialized by sampling from a Beta distribution ($\alpha = 1$ and $\beta = 1$), e was initialized by sampling from a Lognormal distribution ($\mu = 0$, $\sigma = 1$), and all other parameters (α_{1-2} , ρ , κ) were initialized by sampling from a Normal distribution ($\mu = 0$, $\sigma = 1$).

Validation of model fitting procedure

To validate the model fitting procedure, we simulated data using each model and using the model comparison procedure to recover the correct model (Supplementary Table 1). Furthermore, we validated our parameter fits by simulating data using the best fitting parameters for each parameter and then recovering those parameters. The correlations for the parameters of interest were all equal or greater than $r = .74$ and at least .63 for all other parameters (see Supplementary Table 2).

Model comparison

To compare between models, in terms of how well each model accounted for participants' choices, we employed the integrated Bayesian Information Criterion (iBIC)^{39–40}. First, we estimated the evidence for each model (L) by calculating the mean likelihood of the model given 200,000 random parameter settings sampled from the fitted group-level prior distributions.

Subsequently, we computed the iBIC by penalizing the model evidence to account for model complexity, following this formula: $iBIC = 2 \ln L + k \ln n$, where k represents the number of fitted parameters, and n is the number of subject choices used to compute the likelihood. Lower iBIC values indicate a more parsimonious and better-fitting model.

Model simulations

To examine the predictions of the 'controllability' and 'elastic controllability' models, we simulated choices using the best-fitting parameter settings obtained from participants' actual choices. For generality, planets were randomized in the simulation as they were in the actual experiment, such that a simulated participant did not visit precisely the same planets as the actual participant. Furthermore, to avoid discrepancies between models, each planet was repeated the same number of times for both models. 2000 simulations were performed with each model.

Individual differences

We examined individual biases in elasticity inference using participants' best-fitting parameters for beliefs about controllability ($\gamma_{\text{controllability}}$) and elasticity ($\gamma_{\text{elasticity}}$). In addition to these focal parameters, we also examined four other parameters that directly influence opt-in decisions and extra ticket purchases: an inverse temperature/noise parameter (β) and three baseline preference parameters ($\alpha_1, \alpha_2, \alpha_3$).

We then regressed participants' actual opt-in (logistic regression) and extra ticket purchases (probit regression) in each of the 3 controllability conditions on participants' parameters.

Regressions were performed in the lme4 package in R using Rstudio⁴¹. Statistical significance was assessed using the Wald Z Test³⁴ with normality assessed using the Shapiro-Wilk Test³⁵.

Self-report measures

To assess trait-level agency beliefs and related psychopathologies, we administered the Sense of Agency Scale (SOA)⁴². Additionally, we measured dysfunctional attitudes toward problem-solving via the Negative Problem Orientation Questionnaire (NPOQ)⁴³, and temperamental differences (five domains: hyperthymic, dysthymic, cyclothymic, irritable, and anxious) via the Temperament Evaluation Questionnaire (TEMPS-A)^{44–45}.

Additionally, we incorporated a selection of questionnaires used by Gillan et al. (2016)⁴⁶, which assess three factors—Anxious-depression, Compulsive behavior and intrusive thought, and Social withdrawal—linked to goal-directed control deficits. These include the Apathy Evaluation Scale⁴⁷ (AES), Eating Attitudes Test⁴⁸ (EAT), Alcohol Use Disorders Identification Test⁴⁹ (AUDIT), Barratt Impulsiveness Scale⁵⁰ (BIS), Liebowitz Social Anxiety Scale⁵¹ (LSAS), Self-Rating Depression Scale⁵² (SDS), Trait Subscale of the State-Trait Anxiety Inventory for Adults⁵³ (STAI; Spielberger, 1983), and the revised Obsessive-Compulsive Inventory⁵⁴ (OCI-R). We utilized abbreviated versions of these questionnaires following the methodology of Wise and Dolan (2020)⁵⁵, who refined the questionnaires using a classifier to retain the items most contributory to variance in the three factors without significant information loss.

As preregistered, we aggregated the scores of the selected items from each questionnaire independently for inclusion in our CCA. Questionnaires were scored following the corresponding scoring guide, but in such a way that larger scores always indicate greater psychopathology. Based on best practice recommendations^{56–57}, participants were screened for inattentive responding with infrequency items ('I competed in the 1917 Summer Olympic Games') which resulted in the removal of 11 participants (Initial: 6, Replication: 5).

Directly using the scores from questionnaires of varying lengths in the CCA can give undue influence to short questionnaires, whose scores are likely more noisy. To mitigate this problem, we adjusted the standard deviation of each questionnaire's score proportionally to the square root of its number of items before conducting the CCA (in accordance with the formula for standard error). After performing the CCA on these weighted scores and normalized model parameters, the resulting canonical loadings were transformed back to their original standardized scale, ensuring that the reported loadings accurately reflected the original, unweighted scale of the questionnaire scores.

Canonical Correlation Analysis (CCA)

Canonical correlation analysis was implemented using the robust CCA method based on projection pursuit ('ccAPP' package);⁵⁸ We searched for the single canonical dimension that maximizes the Spearman correlation between the model parameters and psychopathology scores. Significance testing was conducted using a two-tailed permutation test with 1,000 iterations. In each iteration, the model parameters were shuffled relative to the psychopathology scores, and the canonical correlation was recalculated. The observed canonical correlation was then compared to this null distribution to obtain a p-value. To evaluate the contribution of the elasticity prior parameter ($\gamma_{\text{elasticity}}$) to the parameter-psychopathology correlation, we performed an additional permutation test. In this test, only the elasticity bias estimates ($\gamma_{\text{elasticity}}$) were shuffled, while all other model parameters and self-report scores remained constant.

To assess how the identified psychopathological profile manifests in raw task measures, we calculated a composite psychopathology score for each participant. This score was derived by multiplying individual participant scores by CCA-derived loadings, representing their position on the psychopathology spectrum. We then used this composite score as a predictor ($\beta_{\text{composite}}$) in a linear regression model to forecast the difference in average opt-in and extra ticket behavior between conditions where controllability was high and inelastic versus high and elastic.

Supplementary Information

Supplementary Note 1: Higher elasticity does not imply higher controllability

Here, we show using two established definitions of controllability that whether an environment is more or less elastic does not determine whether it is more or less controllable.

Definition 1, reward-based controllability¹⁵

If control is defined as the fraction of available reward that is controllably achievable, and we assume all participants are in principle willing and able to invest 3 tickets, controllability can be computed in the present task as:

$$\chi = \text{Max}_{A,C} P(S' = \text{goal} | S, A, C) - \text{Min}_{A,C} P(S' = \text{goal} | S, A, C)$$

where $P(S' = \text{goal} | S, A, C)$ is the probability of reaching the treasure from present state S when taking action A and investing C resources in executing the action. In any of the task environments, the probability of reaching the goal is maximized by purchasing 3 tickets ($C = 3$) and choosing the vehicle that leads to the goal ($A = \text{correct vehicle}$). Conversely, the probability of reaching the goal is minimized by purchasing 3 tickets ($C = 3$) and choosing the vehicle that does not lead to the goal ($A = \text{wrong vehicle}$). This calculation is thus entirely independent of elasticity, since it only considers what would be achieved by maximal resource investment, whereas elasticity consists of

the reduction in controllability that would arise if the maximal available C is reduced. Consequently, any environment where the maximum available control is higher yet varies less with resource investment would be more controllable and less elastic.

Note that if we also account for ticket costs in calculating reward, this will only reduce the fraction of achievable reward and thus the calculated control in elastic environments.

Definition 2, information-theoretic controllability¹[↗](#)²[↗](#)

Here controllability is defined as the reduction in outcome entropy due to knowing which action is taken:

$$I(S'; A, C | S) = H(S'|S) - H(S'|S, A, C)$$

where $H(S'|S)$ is the conditional entropy of the distribution of outcomes S' given the present state

S , and $H(S'|S, A, C)$ is the conditional entropy of the outcome given the present state, action, and resource investment.

To compare controllability, we consider two environments with the same maximum control:

- Inelastic environment: If the correct vehicle is chosen, there is a 100% chance of reaching the goal state with 1, 2, or 3 tickets. Thus, out of 7 possible action-resource investment combinations, three deterministically lead to the goal state (≥ 1 tickets and correct vehicle choice), three never lead to it (≥ 1 tickets and wrong vehicle choice), and one (0 tickets) leads to it 20% of the time (since walking leads to the treasure on 20% of trials).
- Elastic Environment: If the correct vehicle is chosen, the probability of boarding it is 0% with 1 ticket, 50% with 2 tickets, and 100% with 3 tickets. Thus, out of 7 possible action-resource investment combinations, one deterministically leads to the goal state (3 tickets and correct vehicle choice), one never leads to it (3 tickets and wrong vehicle choice), one leads to it 60% of the time (2 tickets and correct vehicle choice: 50% boarding + 50% $\times$ 20% when failing to board), one leads to it 10% of time (2 ticket and wrong vehicle choice), and three lead to it 20% of time (0-1 tickets).

Here we assume a uniform prior over actions, which renders the information-theoretic definition of controllability equal to another definition termed ‘instrumental divergence’^{60,61}. We note that changing the uniform prior assumption would change the results for the two environments, but that would not change the general conclusion that there can be environments that are more controllable yet less elastic.

Step 1: Calculating $H(S'|S)$

For the inelastic environment:

$$P(\text{goal}) = (3 \times 100\% + 3 \times 0\% + 1 \times 20\%)/7 = .46, P(\text{non-goal}) = .54$$

$$H(S'|S) = -[.46 \times \log_2(.46) + .54 \times \log_2(.54)] = 1 \text{ bit}$$

For the elastic environment:

$$P(\text{goal}) = (1 \times 100\% + 1 \times 0\% + 1 \times 60\% + 1 \times 10\% + 3 \times 20\%)/7 = .33, P(\text{non-goal}) = .67$$

$$H(S'|S) = -[.33 \times \log_2(.33) + .67 \times \log_2(.67)] = .91 \text{ bits}$$

Step 2: Calculating $H(S'|S, A, C)$

Inelastic environment: Six action-resource investment combinations have deterministic outcomes entailing zero entropy, whereas investing 0 tickets has a probabilistic outcome (20%). The entropy for 0 tickets is: $H(S'|C=0) = -[.2 \times \log_2(.2) + 0.8 \times \log_2(.8)] = .72$ bits. Since this action- resource investment combination is chosen with probability $1/7$, the total conditional entropy is approximately .10 bits

Elastic environment: 2 actions have deterministic outcomes (3 tickets with correct/wrong vehicle), whereas the other 5 actions have probabilistic outcomes:

- tickets and correct vehicle (60% success):
- $H(S'|A = \text{correct}, C = 2) = -[.6 \times \log_2(.6) + .4 \times \log_2(.4)] = .97$ bits
- tickets and wrong vehicle (10% success):
- $H(S'|A = \text{wrong}, C = 2) = -[.1 \times \log_2(.1) + .9 \times \log_2(.9)] = .47$ bits
- 0-1 tickets (20% success):
- $H(S'|C = 0-1) = -[.2 \times \log_2(.2) + .8 \times \log_2(.8)] = .72$ bits

Thus the total conditional entropy of the elastic environment is: $H(S'|S, A, C) = (1/7) \times .97 + (1/7) \times .47 + (3/7) \times .72 = .52$ bits

Step 3: Calculating $I(S' | A, S)$

Inelastic environment: $I(S'; A, C | S) = H(S'|S) - H(S'|S, A, C) = 1 - 0.1 = .9$ bits

Elastic environment: $I(S'; A, C | S) = H(S'|S) - H(S'|S, A, C) = .91 - .52 = .39$ bits

Thus, the inelastic environment offers higher information-theoretic controllability (.9 bits) compared to the elastic environment (.39 bits).

Of note, even if each combination of cost and success/failure to reach the goal is defined as a distinct outcome, then information-theoretic controllability is higher for the inelastic (2.81 bits) than for the elastic (2.30 bits) environment.

Supplementary Table 1.

Model validation.

10 full experimental datasets were simulated using each model. Rows indicate the model used to simulate data and columns indicate the model recovered from the data using the model comparison procedure.

	Controllability	Elastic controllability	Null
Controllability	10	0	0
Elastic controllability	0	10	0
Null	0	0	10

Supplementary Table 2.

Parameter Recovery.

We validated our model fitting procedure by simulating data using the best fitting parameters for each subject and then recovering those parameters. Our correlation between simulated and recovered parameters was at least .74 for all parameters of interest that capture the effects of the experimental conditions, and at least .59 for all other parameters.

Parameter	Correlation between Simulated and Recovered Parameters
β	.91
$\lambda_{\text{controllability}}$	.72
$\epsilon_{\text{controllability}}$	.63
$\gamma_{\text{controllability}}$	.77
$\lambda_{\text{elasticity}}$	.81
$\epsilon_{\text{elasticity}}$	.59
$\gamma_{\text{elasticity}}$	.74
α_1	.82
α_2	.81
α_3	.81
ρ	.89
κ	.63

Supplementary Table 3.

Model parameters.

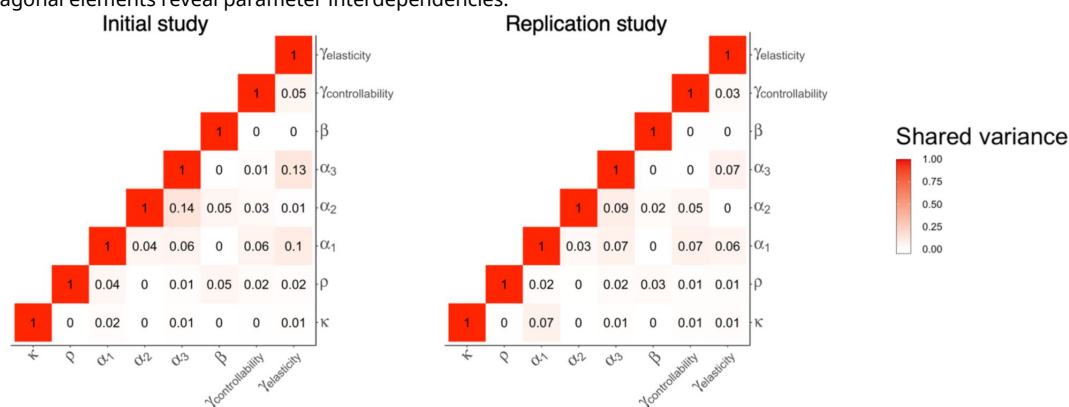
The table presents means, standard deviations, prior distributions, and interpretations for each parameter of the three computational models. Effective controllability and elasticity biases are computed as $\gamma = (\lambda - 0.5) \times \epsilon$. κ is only included in the modified elastic controllability* model.

Elastic controllability model				
Parameter	Mean	SD	Prior	Interpretation
β	0.07	0.02	$\beta(1,1)$	SoftMax decision noise
$\lambda_{\text{controllability}}$	.69	.17	$\beta(1,1)$	Controllability bias
$\epsilon_{\text{controllability}}$	19.9	20	$\exp(N(0,1))$	Controllability bias strength
$\lambda_{\text{elasticity}}$	.16	.14	$\beta(1,1)$	Elasticity bias
$\epsilon_{\text{elasticity}}$	.61	.20	$\exp(N(0,1))$	Elasticity bias strength
ρ	2.2	0.66	$N(0,1)$	Choice perseveration
α_1	1.3	0.28	$N(0,1)$	Fixed preference for purchasing 1 ticket
α_2	0.63	0.37	$N(0,1)$	Fixed preference for purchasing 2 tickets
α_3	-0.74	0.37	$N(0,1)$	Fixed preference for purchasing 3 tickets
κ	0.13	0.11	$N(0,1)$	Prioritize resolving max controllability upon failing with 3 tickets
Controllability model				
Parameter	Mean	SD	Prior	Interpretation
β	0.06	0.01	$\beta(1,1)$	SoftMax decision noise
$\lambda_1 \text{ ticket}$	.65	.20	$\beta(1,1)$	1 ticket bias
$\epsilon_1 \text{ ticket}$	.36	.42	$\exp(N(0,1))$	1 ticket bias strength concentration
$\lambda_2 \text{ tickets}$	.52	.12	$\beta(1,1)$	2 ticket bias
$\epsilon_2 \text{ tickets}$	.88	.52	$\exp(N(0,1))$	2 ticket bias strength concentration
$\lambda_3 \text{ tickets}$	.11	.04	$\beta(1,1)$	3 ticket bias
$\epsilon_3 \text{ tickets}$	7.97	.91	$\exp(N(0,1))$	3 ticket bias strength concentration
ρ	2.03	0.64	$N(0,1)$	Choice perseveration
α_1	.25	0.58	$N(0,1)$	Fixed preference for purchasing 1 ticket
α_2	0.47	0.57	$N(0,1)$	Fixed preference for purchasing 2 tickets
α_3	1.78	0.59	$N(0,1)$	Fixed preference for purchasing 3 tickets
Null model				
Parameter	Mean	SD	Prior	Interpretation
ρ	2.34	0.59	$N(0,1)$	Choice perseveration
α_1	.19	0.35	$N(0,1)$	Fixed preference for purchasing 1 ticket
α_2	.05	0.41	$N(0,1)$	Fixed preference for purchasing 2 tickets
α_3	-.33	.75	$N(0,1)$	Fixed preference for purchasing 3 tickets

Supplementary Figure 1.

Shared variance between elastic controllability* model parameters.

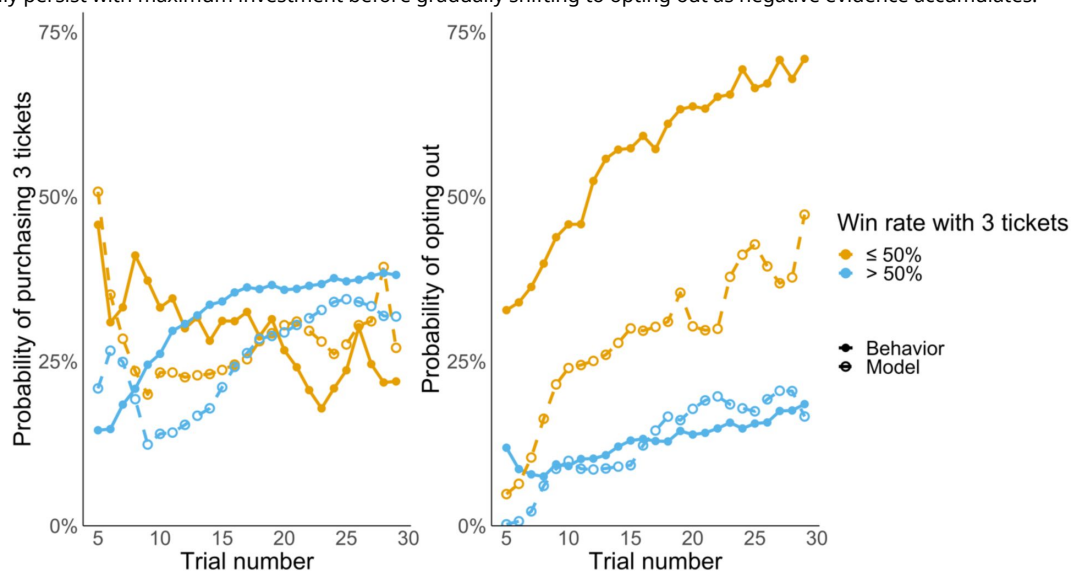
Heatmaps display squared Pearson correlations between parameter pairs for initial (left) and replication (right) studies. Color intensity indicates shared variance magnitude (0-1). Parameters: $\gamma_{\text{controllability}}$, $\gamma_{\text{elasticity}}$, β , α_{1-3} , ρ , κ . Diagonal elements=1; off-diagonal elements reveal parameter interdependencies.

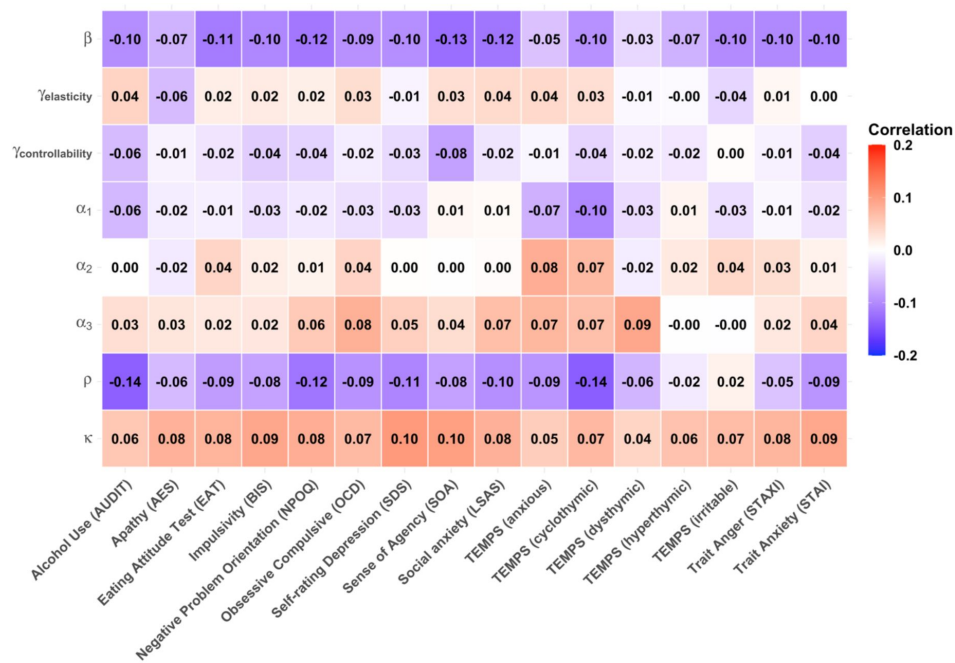


Supplementary Figure 2.

Behavioral trajectories under high and low controllability conditions. Left:

Probability of investing maximum resources (3 tickets) across trials. **Right:** Probability of opting out. Data points are divided by experienced win rates with 3 tickets (orange: $\leq 50\%$, blue: $> 50\%$). Solid lines with filled circles represent participant behavior; dashed lines with empty circles show computational model predictions. Participants experiencing low controllability initially persist with maximum investment before gradually shifting to opting out as negative evidence accumulates.

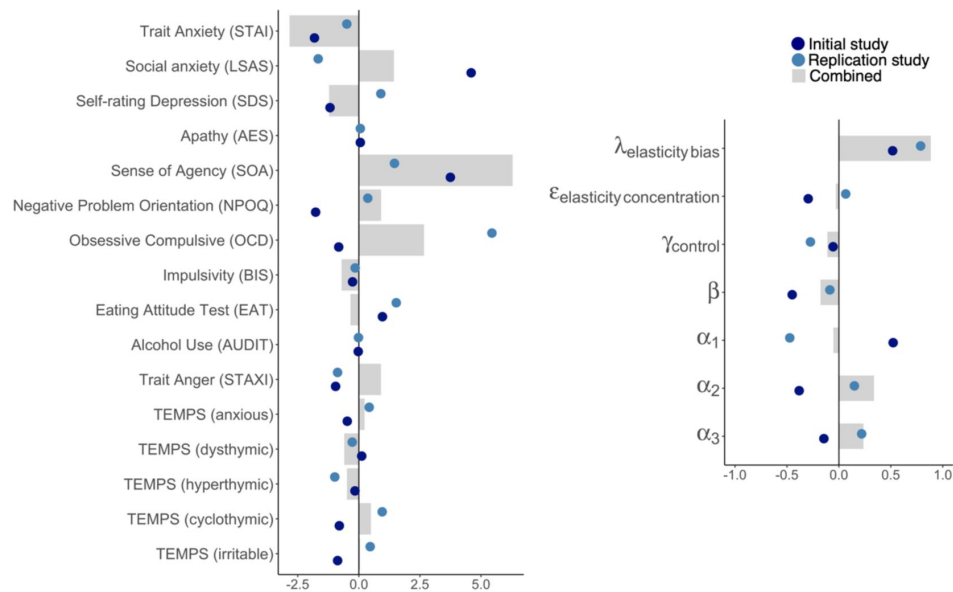




Supplementary Figure 3.

Correlation coefficients between model parameter and questionnaire scores.

Pearson correlation of computational model parameters (rows) with psychological questionnaire scores (columns). Color intensity indicates correlation strength, with blue representing negative correlations and red representing positive correlations. No correlation reached statistical significance when correcting for multiple comparisons (FDR).



Supplementary Figure 4.

Model parameters and psychopathology.

Loadings from a Canonical Correlation Analysis (CCA) between model parameters and self-report psychopathology measures while separating concentration from bias parameters in the elasticity prior. Self-report measures were scored such that higher values reflect higher levels of psychopathology. Loadings are shown for the Initial (dark blue), Replication (light blue), and combined (gray bars) datasets. The magnitude of the loading reflects the degree to which each measure contributes to the correlation between parameters and self-reports

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Reviewer #1 (Public review):

Summary:

The authors investigated the elasticity of controllability by developing a task that manipulates the probability of achieving a goal with a baseline investment (which they refer to as inelastic controllability) and the probability that additional investment would increase the probability of achieving a goal (which they refer to as elastic controllability). They found that a computational model representing the controllability and elasticity of the environment accounted better for the data than a model representing only the controllability. They also found that prior biases about the controllability and elasticity of the environment was associated with a composite psychopathology score. The authors conclude that elasticity inference and bias guide resource allocation.

Strengths:

This research takes a novel theoretical and methodological approach to understanding how people estimate the level of control they have over their environment and how they adjust their actions accordingly. The task is innovative and both it and the findings are well-described (with excellent visuals). They also offer thorough validation for the particular model they develop. The research has the potential to theoretically inform understanding of control across domains, which is a topic of great importance.

Weaknesses:

In its revised form, the manuscript addresses most of my previous concerns. The main remaining weakness pertains to the analyses aimed at addressing my suggesting of Bayesian updating as an alternative to the model proposed by the authors. My suggestion was to assume that people perform a form of function approximation to relate resource expenditure to success probability. The authors performed a version of this where people were weighing evidence for a few canonical functions (flat, step, linear), and found that this model underperformed theirs. However, this Bayesian model is quite constrained in its ability to estimate the function relating resources. A more robust test would be to assume a more flexible form of updating that is able to capture a wide range of distributions (e.g., using basis functions, gaussian processes, or nonparametric estimators); see, e.g., work by Griffiths on human function learning). The benefit of testing this type of model is that it would make contact with a known form of inference that individuals engage in across various settings and therefore could offer a more parsimonious and generalizable account of function learning, whereby learning of resource elasticity is a special case. I defer to the authors as to whether they'd like to pursue this direction, but if not I think it's still important that they acknowledge that they are unable to rule out a more general process like this as an alternative to their model. This pertains also to inferences about individual differences, which currently hinge on their preferred model being the most parsimonious.

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Reviewer #2 (Public review):**Summary:**

In this paper, the authors test whether controllability beliefs and associated actions/resource allocation are modulated by things like time, effort, and monetary costs (what they call "elastic" as opposed to "inelastic" controllability). Using a novel behavioral task and computational modeling, they find that participants do indeed modulate their resources depending on whether they are in an "elastic," "inelastic," or "low controllability" environment. The authors also find evidence that psychopathology is related to specific biases in controllability.

Strengths:

This research investigates how people might value different factors that contribute to controllability in a creative and thorough way. The authors use computational modeling to try to dissociate "elasticity" from "overall controllability," and find some differential associations with psychopathology. This was a convincing justification for using modeling above and beyond behavioral output and yielded interesting results. Notably, the authors conclude that these findings suggest that biased elasticity could distort agency beliefs via maladaptive resource allocation. Overall, this paper reveals important findings about how people consider components of controllability.

Weaknesses:

The authors have gone to great lengths to revise the manuscript to clarify their definitions of "elastic" and "inelastic" and bolster evidence for their computational model, resulting in an overall strong manuscript that is valuable for elucidating controllability dynamics and preferences. One minor weakness is that the justification for the analysis technique for the relationships between the model parameters and the psychopathology measures remains lacking given the fact that simple correlational analyses did not reveal any significant associations.

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Reviewer #3 (Public review):

A bias in how people infer the amount of control they have over their environment is widely believed to be a key component of several mental illnesses including depression, anxiety, and addiction. Accordingly, this bias has been a major focus in computational models of those disorders. However, all of these models treat control as a unidimensional property, roughly, how strongly outcomes depend on action. This paper proposes—correctly, I think—that the intuitive notion of "control" captures multiple dimensions in the relationship between action and outcome. In particular, the authors identify one key dimension: the degree to which outcome depends on how much *effort* we exert, calling this dimension the "elasticity of control". They additionally argue that this dimension (rather than the more holistic notion of controllability) may be specifically impaired in certain types of psychopathology. This idea has the potential to change how we think about several major mental disorders in a substantial way and can additionally help us better understand how healthy people navigate challenging decision-making problems. More concisely, it is a very good idea.

Unfortunately, my view is that neither the theoretical nor empirical aspects of the paper really deliver on that promise. In particular, most (perhaps all) of the interesting claims in the paper have weak empirical support.

Starting with theory, the authors do not provide a strong formal characterization of the proposed notion of elasticity. There are existing, highly general models of controllability (e.g., Huys & Dayan, 2009; Ligneul, 2021) and the elasticity idea could naturally be embedded within one of these frameworks. The authors gesture at this in the introduction; however, this formalization is not reflected in the implemented model, which is highly task-specific. Moreover, the authors present elasticity as if it is somehow "outside of" the more general notion of controllability. However, effort and investment are just specific dimensions of action; and resources like money, strength, and skill (the "highly trained birke") are just specific dimensions of state. Accordingly, the notion of elasticity is necessarily implicitly captured by the standard model. Personally, I am compelled by the idea that effort and resource (and therefore elasticity) are particularly important dimensions, ones that people are uniquely tuned to. However, by framing elasticity as a property that is different in kind from controllability (rather than just a dimension of controllability), the authors only make it more difficult to integrate this exciting idea into generalizable models.

Turning to experiment, the authors make two key claims: (1) people infer the elasticity of control, and (2) individual differences in how people make this inference are importantly related to psychopathology.

Starting with claim 1, there are three subclaims here; implicitly, the authors make all three. (1A) People's behavior is sensitive to differences in elasticity, (1B) people actually represent/track something like elasticity, and (1C) people do so naturally as they go about

their daily lives. The results clearly support 1A. However, 1B and 1C are not strongly supported.

(1B) The experiment cannot support the claim that people represent or track elasticity because effort is the only dimension over which participants can engage in any meaningful decision-making. The other dimension, selecting which destination to visit, simply amounts to selecting the location where you were just told the treasure lies. Thus, any adaptive behavior will necessarily come out in a sensitivity to how outcomes depend on effort.

Notes on rebuttal: The argument that vehicle/destination choice is not trivial because people occasionally didn't choose the instructed location is not compelling to me—if anything, the exclusion rate is unusually low for online studies. The finding that people learn more from non-random outcomes is helpful, but this could easily be cast as standard model-based learning very much like what one measures with the Daw two-step task (nothing specific to control here). Their final argument is the strongest, that to explain behavior the model must assume "a priori that increased effort could enhance control." However, more literally, the necessary assumption is that each attempt increases the probability of success—e.g. you're more likely to get a heads in two flips than one. I suppose you can call that "elasticity inference", but I would call it basic probabilistic reasoning.

For 1C, the claim that people infer elasticity outside of the experimental task cannot be supported because the authors explicitly tell people about the two notions of control as part of the training phase: "To reinforce participants' understanding of how elasticity and controllability were manifested in each planet, [participants] were informed of the planet type they had visited after every 15 trips." (line 384).

Notes on rebuttal: The authors try to retreat, saying "our research question was whether people can distinguish between elastic and inelastic controllability." I struggle to reconcile this with the claim in the abstract "These findings establish the elasticity of control as a distinct cognitive construct guiding adaptive behavior". That claim is the interesting one, and the one I am evaluating the evidence in light of.

Finally, I turn to claim 2, that individual differences in how people infer elasticity are importantly related to psychopathology. There is much to say about the decision to treat psychopathology as a unidimensional construct (the authors claim otherwise, but see Fig 6C). However, I will keep it concrete and simply note that CCA (by design) obscures the relationship between any two variables. Thus, as suggestive as Figure 6B is, we cannot conclude that there is a strong relationship between Sense of Agency (SOA) and the elasticity bias—this result is consistent with any possible relationship (even a negative one). As it turns out, Figure S3 shows that there is effectively no relationship ($r=0.03$).

Notes on rebuttal: The authors argue for CCA by appeal to the need to "account for the substantial variance that is typically shared among different forms of psychopathology". I agree. A simple correlation would indeed be fairly weak evidence. Strong evidence would show a significant correlation after *controlling for* other factors (e.g. a regression predicting elasticity bias from all subscales simultaneously). CCA effectively does the opposite, asking whether—with the help of all the parameters and all the surveys—one can find any correlation between the two sets of variables. The results are certainly suggestive, but they provide very little statistical evidence that the elasticity parameter is meaningfully related to any particular dimension of psychopathology.

There is also a feature of the task that limits our ability to draw strong conclusions about individual differences about elasticity inference. In the original submission, the authors stated that the study was designed to be "especially sensitive to overestimation of elasticity". A straightforward consequence of this is that the resulting *empirical* estimate of estimation bias (i.e., the $\gamma_{\text{elasticity}}$ parameter) is itself biased. This immediately undermines any

claim that references the directionality of the elasticity bias (e.g. in the abstract). Concretely, an undirected deficit such as slower learning of elasticity would appear as a directed overestimation bias.

When we further consider that elasticity inference is the only meaningful learning/decision-making problem in the task (argued above), the situation becomes much worse. Many general deficits in learning or decision-making would be captured by the elasticity bias parameter. Thus, a conservative interpretation of the results is simply that psychopathology is associated with impaired learning and decision-making.

Notes on rebuttal: I am very concerned to see that the authors removed the discussion of this limitation in response to my first review. I quote the original explanation here:

- In interpreting the present findings, it needs to be noted that we designed our task to be especially sensitive to overestimation of elasticity. We did so by giving participants free 3 tickets at their initial visits to each planet, which meant that upon success with 3 tickets, people who overestimate elasticity were more likely to continue purchasing extra tickets unnecessarily. Following the same logic, had we first had participants experience 1 ticket trips, this could have increased the sensitivity of our task to underestimation of elasticity in elastic environments. Such underestimation could potentially relate to a distinct psychopathological profile that more heavily loads on depressive symptoms. Thus, by altering the initial exposure, future studies could disambiguate the dissociable contributions of overestimating versus underestimating elasticity to different forms of psychopathology.

The logic of this paragraph makes perfect sense to me. If you assume low elasticity, you will infer that you could catch the train with just one ticket. However, when elasticity is in fact high, you would find that you don't catch the train, leading you to quickly infer high elasticity-eliminating the bias. In contrast, if you assume high elasticity, you will continue purchasing three tickets and will never have the opportunity to learn that you could be purchasing only one-the bias remains.

The authors attempt to argue that this isn't happening using parameter recovery. However, they only report the **correlation** in the parameter, whereas the critical measure is the **bias**. Furthermore, in parameter recovery, the data-generating and data-fitting models are identical-this will yield the best possible recovery results. Although finding no bias in this setting would support the claims, it cannot outweigh the logical argument for the bias that they originally laid out. Finally, parameter recovery should be performed across the full range of plausible parameter values; using fitted parameters (a detail I could only determine by reading the code) yields biased results because the fitted parameters are themselves subject to the bias (if present). That is, if true low elasticity is inferred as high elasticity, then you will not have any examples of low elasticity in the fitted parameters and will not detect the inability to recover them.

Minor comments:

Below are things to keep in mind.

The statistical structure of the task is inconsistent with the framing. In the framing, participants can make either one or two second boarding attempts (jumps) by purchasing extra tickets. The additional attempt(s) will thus succeed with probability p for one ticket and $2p - p^2$ for two tickets; the p^2 captures the fact that you only take the second attempt if you fail on the first. A consequence of this is buying more tickets has diminishing returns. In contrast, in the task, participants always jumped twice after purchasing two tickets, and the probability of success with two tickets was exactly double that with one ticket. Thus, if participants are applying an intuitive causal model to the task, they will appear to "underestimate" the elasticity of control. I don't think this seriously jeopardizes the key

results, but any follow-up work should ensure that the task's structure is consistent with the intuitive causal model.

The model is heuristically defined and does not reflect Bayesian updating. For example, it over-estimates maximum control by not using losses with less than 3 tickets (intuitively, the inference here depends on what your beliefs about elasticity). Including forced three-ticket trials at the beginning of each round makes this less of an issue; but if you want to remove those trials, you might need to adjust the model. The need to introduce the modified model with kappa is likely another symptom of the heuristic nature of the model updating equations.

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Author response:

The following is the authors' response to the original reviews

We thank the Reviewers for their thorough reading and thoughtful feedback. Below, we address each of the concerns raised in the public reviews, and outline our revisions that aim to further clarify and strengthen the manuscript.

In our response, we clarify our conceptualization of elasticity as a dimension of controllability, formalizing it within an information-theoretic framework, and demonstrating that controllability and its elasticity are partially dissociable. Furthermore, we provide clarifications and additional modeling results showing that our experimental design and modeling approach are well-suited to dissociating elasticity inference from more general learning processes, and are not inherently biased to find overestimates of elasticity. Finally, we clarify the advantages and disadvantages of our canonical correlation analysis (CCA) approach for identifying latent relationships between multidimensional data sets, and provide additional analyses that strengthen the link between elasticity estimation biases and a specific psychopathology profile.

Public Reviews:

Reviewer 1 (Public review):

This research takes a novel theoretical and methodological approach to understanding how people estimate the level of control they have over their environment, and how they adjust their actions accordingly. The task is innovative and both it and the findings are well-described (with excellent visuals). They also offer thorough validation for the particular model they develop. The research has the potential to theoretically inform the understanding of control across domains, which is a topic of great importance.

We thank the Reviewer for their favorable appraisal and valuable suggestions, which have helped clarify and strengthen the study's conclusion.

An overarching concern is that this paper is framed as addressing resource investments across domains that include time, money, and effort, and the introductory examples focus heavily on effort-based resources (e.g., exercising, studying, practicing). The experiments, though, focus entirely on the equivalent of monetary resources - participants make discrete actions based on the number of points they want to use on a given turn. While the same ideas might generalize to decisions about other kinds of resources (e.g., if participants were having to invest the effort to reach a goal), this seems like the kind of speculation that would be better reserved for the Discussion section

rather than using effort investment as a means of introducing a new concept (elasticity of control) that the paper will go on to test.

We thank the Reviewer for pointing out a lack of clarity regarding the kinds of resources tested in the present experiment. Investing additional resources in the form of extra tickets did not only require participants to pay more money. It also required them to invest additional time – since each additional ticket meant making another attempt to board the vehicle, extending the duration of the trial, and attentional effort – since every attempt required precisely timing a spacebar press as the vehicle crossed the screen. Given this involvement of money, time, and effort resources, we believe it would be imprecise to present the study as concerning monetary resources in particular. That said, we agree with the Reviewer that results might differ depending on the resource type that the experiment or the participant considers most. Thus, we now clarify the kinds of resources the experiment involved (lines 87-97):

“To investigate how people learn the elasticity of control, we allowed participants to invest different amounts of resources in attempting to board their preferred vehicle. Participants could purchase one (40 coins), two (60 coins), or three tickets (80 coins) or otherwise walk for free to the nearest location. Participants were informed that a single ticket allowed them to board only if the vehicle stopped at the station, while additional tickets provided extra chances to board even after the vehicle had left the platform. For each additional ticket, the chosen vehicle appeared moving from left to right across the screen, and participants could attempt to board it by pressing the spacebar when it reached the center of the screen. Thus, each additional ticket could increase the chance of boarding but also required a greater investment of resources—decreasing earnings, extending the trial duration, and demanding attentional effort to precisely time a button press when attempting to board.”

In addition, in the revised discussion, we now highlight the open question of whether inferences concerning the elasticity of control generalize across different resource domains (lines 341-348):

“Another interesting possibility is that individual elasticity biases vary across different resource types (e.g., money, time, effort). For instance, a given individual may assume that controllability tends to be highly elastic to money but inelastic to effort. Although the task incorporated multiple resource types (money, time, and attentional effort), the results may differ depending on the type of resources on which the participant focuses. Future studies could explore this possibility by developing tasks that separately manipulate elasticity with respect to different resource types. This would clarify whether elasticity biases are domain-specific or domain-general, and thus elucidate their impact on everyday decision-making.”

Setting aside the framing of the core concepts, my understanding of the task is that it effectively captures people's estimates of the likelihood of achieving their goal ($\Pr(\text{success})$) conditional on a given investment of resources. The ground truth across the different environments varies such that this function is sometimes flat (low controllability), sometimes increases linearly (elastic controllability), and sometimes increases as a step function (inelastic controllability). If this is accurate, then it raises two questions.

First, on the modeling front, I wonder if a suitable alternative to the current model would be to assume that the participants are simply considering different continuous functions like these and, within a Bayesian framework, evaluating the probabilistic evidence for each function based on each trial's outcome. This would give participants an estimate of the marginal increase in $\Pr(\text{success})$ for each ticket, and they could then weigh the expected value of that ticket choice ($\Pr(\text{success}) \times 150$ points) against the marginal increase in point cost for each ticket. This should yield similar predictions for optimal

performance (e.g., opt-out for lower controllability environments, i.e., flatter functions), and the continuous nature of this form of function approximation also has the benefit of enabling tests of generalization to predict changes in behavior if there was, for instance, changes in available tickets for purchase (e.g., up to 4 or 5) or changes in ticket prices. Such a model would of course also maintain a critical role for priors based on one's experience within the task as well as over longer timescales, and could be meaningfully interpreted as such (e.g., priors related to the likelihood of success/failure and whether one's actions influence these). It could also potentially reduce the complexity of the model by replacing controllability-specific parameters with multiple candidate functions (presumably learned through past experience, and/or tuned by experience in this task environment), each of which is being updated simultaneously.

We thank the Reviewer for suggesting this interesting alternative modeling approach. We agree that a Bayesian framework evaluating different continuous functions could offer advantages, particularly in its ability to generalize to other ticket quantities and prices. To test the Reviewer's suggestion, we implemented a Bayesian model where participants continuously estimate both controllability and its elasticity as a mixture of three archetypal functions mapping ticket quantities to success probabilities. The flat function provides no control regardless of how many tickets are purchased (corresponding to low controllability). The step function provides the same level of control as long as at least one ticket is purchased (inelastic controllability). The linear function increases control proportionally with each additional ticket (elastic controllability). The model computes the likelihood that each of the functions produced each new observation, and accordingly updates its beliefs. Using these beliefs, the model estimates the probability of success for purchasing each number of tickets, allowing participants to weigh expected control against increasing ticket costs. Despite its theoretical advantages for generalization to different ticket quantities, this continuous function approximation model performed significantly worse than our elastic controllability model (log Bayes Factor > 4100 on combined datasets). We surmise that the main advantage offered by the elastic controllability model is that it does not assume a linear increase in control as a function of resource investment – even though this linear relationship was actually true in our experiment and is required for generalizing to other ticket quantities, it likely does not match what participants were doing. We present these findings in a new section ‘Testing alternative methods’ (lines 686-701):

“We next examined whether participant behavior would be better characterized as a continuous function approximation rather than the discrete inferences in our model. To test this, we implemented a Bayesian model where participants continuously estimate both controllability and its elasticity as a mixture of three archetypal functions mapping ticket quantities to success probabilities. The flat function provides no control regardless of how many tickets are purchased (corresponding to low controllability). The step function provides full control as long as at least one ticket is purchased (inelastic controllability). The linear function linearly increases control with the number of extra tickets (i.e., 0%, 50%, and 100% control for 1, 2, and 3 tickets, respectively; elastic controllability). The model computes the likelihood that each of the functions produced each new observation, and accordingly updates its beliefs. Using these beliefs, the model estimates the probability of success for purchasing each number of tickets, allowing participants to weigh expected control against increasing ticket costs. Despite its theoretical advantages for generalization to different ticket quantities, this continuous function approximation model performed significantly worse than the elastic controllability model (log Bayes Factor > 4100 on combined datasets), suggesting that participants did not assume that control increases linearly with resource investment.”

We also refer to this analysis in our updated discussion (326-339):

“Second, future models could enable generalization to levels of resource investment not previously experienced. For example, controllability and its elasticity could be jointly estimated via function approximation that considers control as a function of invested resources. Although our implementation of this model did not fit participants’ choices well (see Methods), other modeling assumptions or experimental designs may offer a better test of this idea.”

Second, if the reframing above is apt (regardless of the best model for implementing it), it seems like the taxonomy being offered by the authors risks a form of "jangle fallacy," in particular by positing distinct constructs (controllability and elasticity) for processes that ultimately comprise aspects of the same process (estimation of the relationship between investment and outcome likelihood). Which of these two frames is used doesn't bear on the rigor of the approach or the strength of the findings, but it does bear on how readers will digest and draw inferences from this work. It is ultimately up to the authors which of these they choose to favor, but I think the paper would benefit from some discussion of a common-process alternative, at least to prevent too strong of inferences about separate processes/modes that may not exist. I personally think the approach and findings in this paper would also be easier to digest under a common-construct approach rather than forcing new terminology but, again, I defer to the authors on this.

We acknowledge the Reviewer's important point about avoiding a potential "jangle fallacy." We entirely agree with the Reviewer that elasticity and controllability inferences are not distinct processes. Specifically, we view resource elasticity as a dimension of controllability, hence the name of our 'elastic controllability' model. In response to this and other Reviewers' comments, in the revised manuscript, we now offer a formal definition of elasticity as the reduction in uncertainty about controllability due to knowing the amount of resources available to the agent (lines 16-20; see further details in response to Reviewer 3 below).

With respect to how this conceptualization is expressed in the modeling, we note that the representation in our model of maximum controllability and its elasticity via different variables is analogous to how a distribution may be represented by separate mean and variance parameters. Even the model suggested by the Reviewer required a dedicated variable representing elastic controllability, namely the probability of the linear controllability function. More generally, a single-process account allows that different aspects of the said process would be differently biased (e.g., one can have an accurate estimate of the mean of a distribution but overestimate its variance). Therefore, our characterization of distinct elasticity and controllability biases (or to put it more accurately, 'elasticity of controllability bias' and 'maximum controllability bias') is consistent with a common construct account.

To avoid misunderstandings, we have now modified the text to clarify that we view elasticity as a dimension of controllability that can only be estimated in conjunction with controllability. Here are a few examples:

Lines 21-28: “While only controllable environments can be elastic, the inverse is not necessarily true – controllability can be high, yet inelastic to invested resources – for example, choosing between bus routes affords equal control over commute time to anyone who can afford the basic fare (Figure 1; Supplementary Note 1). That said, since all actions require some resource investment, no controllable environment is completely inelastic when considering the full spectrum of possible agents, including those with insufficient resources to act (e.g., those unable to purchase a bus fare or pay for a fixed-price meal).”

Lines 45-47: “Experimental paradigms to date have conflated overall controllability and its elasticity, such that controllability was either low or elastic[16-20]. The elasticity of control,

however, must be dissociated from overall controllability to accurately diagnose mismanagement of resources.”

Lines 70-72: “These findings establish elasticity as a crucial dimension of controllability that guides adaptive behavior, and a computational marker of control-related psychopathology.”

Lines 87-88: “To investigate how people learn the elasticity of control, we allowed participants to invest different amounts of resources in attempting to board their preferred vehicle.”

Reviewer 2 (Public review):

This research investigates how people might value different factors that contribute to controllability in a creative and thorough way. The authors use computational modeling to try to dissociate "elasticity" from "overall controllability," and find some differential associations with psychopathology. This was a convincing justification for using modeling above and beyond behavioral output and yielded interesting results. Interestingly, the authors conclude that these findings suggest that biased elasticity could distort agency beliefs via maladaptive resource allocation. Overall, this paper reveals some important findings about how people consider components of controllability.

We appreciate the Reviewer's positive assessment of our findings and computational approach to dissociating elasticity and overall controllability.

The primary weakness of this research is that it is not entirely clear what is meant by "elastic" and "inelastic" and how these constructs differ from existing considerations of various factors/calculations that contribute to perceptions of and decisions about controllability. I think this weakness is primarily an issue of framing, where it's not clear whether elasticity is, in fact, theoretically dissociable from controllability. Instead, it seems that the elements that make up "elasticity" are simply some of the many calculations that contribute to controllability. In other words, an "elastic" environment is inherently more controllable than an "inelastic" one, since both environments might have the same level of predictability, but in an "elastic" environment, one can also partake in additional actions to have additional control over achieving the goal (i.e., expend effort, money, time).

We thank the Reviewer for highlighting the lack of clarity about the concept of elasticity. We first clarify that elasticity cannot be entirely dissociated from controllability because it is a dimension of controllability. If no controllability is afforded, then there cannot be elasticity or inelasticity. This is why in describing the experimental environments, we only label high-controllability, but not low-controllability, environments as ‘elastic’ or ‘inelastic’. For further details on this conceptualization of elasticity, and associated revisions of the text, see our response above to Reviewer 1.

Second, we now clarify that controllability can also be computed without knowing the amount of resources the agent is able and willing to invest, for instance by assuming infinite resources available or a particular distribution of resource availabilities. However, knowing the agent's available resources often reduces uncertainty concerning controllability. This reduction in uncertainty is what we define as elasticity. Since any action requires some resources, this means that no controllable environment is entirely inelastic if we also consider agents that do not have enough resources to commit any action. However, even in this case, environments can differ in the degree to which they are elastic. For further details on this formal definition, and associated revisions of the text, see our response to Reviewer 3.

Importantly, whether an environment is more or less elastic does not fully determine whether it is more or less controllable. In particular, environments can be more controllable yet less elastic. This is true even if we allow that investing different levels of resources (i.e., purchasing 0, 1, 2, or 3 tickets) constitute different actions, in conjunction with participants' vehicle choices. Below, we show this using two existing definitions of controllability.

Definition 1, reward-based controllability[1]: If control is defined as the fraction of available reward that is controllably achievable, and we assume all participants are in principle willing and able to invest 3 tickets, controllability can be computed in the present task as:

$$\chi = \text{Max}_{A,C} P(S' = \text{goal} | S, A, C) - \text{Min}_{A,C} P(S' = \text{goal} | S, A, C)$$

where $P(S' = \text{goal} | S, A, C)$ is the probability of reaching the treasure from present state S when taking action A and investing C resources in executing the action. In any of the task environments, the probability of reaching the goal is maximized by purchasing 3 tickets ($C = 3$) and choosing the vehicle that leads to the goal ($A = \text{correct vehicle}$). Conversely, the probability of reaching the goal is minimized by purchasing 3 tickets ($C = 3$) and choosing the vehicle that does not lead to the goal ($A = \text{wrong vehicle}$). This calculation is thus entirely independent of elasticity, since it only considers what would be achieved by maximal resource investment, whereas elasticity consists of the reduction in controllability that would arise if the maximal available C is reduced. Consequently, any environment where the maximum available control is higher yet varies less with resource investment would be more controllable and less elastic.

Note that if we also account for ticket costs in calculating reward, this will only reduce the fraction of achievable reward and thus the calculated control in elastic environments.

Definition 2, information-theoretic controllability[2]: Here controllability is defined as the reduction in outcome entropy due to knowing which action is taken:

$$I(S'; A, C | S) = H(S' | S) - H(S' | S, A, C)$$

where $H(S' | S)$ is the conditional entropy of the distribution of outcomes S' given the present state S , and $H(S' | S, A, C)$ is the conditional entropy of the outcome given the present state, action, and resource investment.

To compare controllability, we consider two environments with the same maximum control:

- **Inelastic environment:** If the correct vehicle is chosen, there is a 100% chance of reaching the goal state with 1, 2, or 3 tickets. Thus, out of 7 possible action-resource investment combinations, three deterministically lead to the goal state (≥ 1 tickets and correct vehicle choice), three never lead to it (≥ 1 tickets and wrong vehicle choice), and one (0 tickets) leads to it 20% of the time (since walking leads to the treasure on 20% of trials).
- **Elastic Environment:** If the correct vehicle is chosen, the probability of boarding it is 0% with 1 ticket, 50% with 2 tickets, and 100% with 3 tickets. Thus, out of 7 possible action-resource investment combinations, one deterministically leads to the goal state (3 tickets and correct vehicle choice), one never leads to it (3 tickets and wrong vehicle choice), one leads to it 60% of the time (2 tickets and correct vehicle choice: $50\% \text{ boarding} + 50\% \times 20\%$ when failing to board), one leads to it 10% of time (2 ticket and wrong vehicle choice), and three lead to it 20% of time (0-1 tickets).

Here we assume a uniform prior over actions, which renders the information-theoretic definition of controllability equal to another definition termed ‘instrumental divergence’[3,4]. We note that changing the uniform prior assumption would change the results for the two environments, but that would not change the general conclusion that there can be environments that are more controllable yet less elastic.

Step 1: Calculating $H(S'|S)$

For the inelastic environment:

$$P(\text{goal}) = (3 \times 100\% + 3 \times 0\% + 1 \times 20\%)/7 = .46, P(\text{non-goal}) = .54 \quad H(S'|S) = - [.46 \times \log_2(.46) + .54 \times \log_2(.54)] = 1 \text{ bit}$$

For the elastic environment:

$$P(\text{goal}) = (1 \times 100\% + 1 \times 0\% + 1 \times 60\% + 1 \times 10\% + 3 \times 20\%)/7 = .33, P(\text{non-goal}) = .67 \quad H(S'|S) = - [.33 \times \log_2(.33) + .67 \times \log_2(.67)] = .91 \text{ bits}$$

Step 2: Calculating $H(S'|S, A, C)$

Inelastic environment: Six action-resource investment combinations have deterministic outcomes entailing zero entropy, whereas investing 0 tickets has a probabilistic outcome (20%). The entropy for 0 tickets is: $H(S'|C=0) = -[.2 \times \log_2(.2) + 0.8 \times \log_2(.8)] = .72$ bits. Since this action-resource investment combination is chosen with probability 1/7, the total conditional entropy is approximately .10 bits

Elastic environment: 2 actions have deterministic outcomes (3 tickets with correct/wrong vehicle), whereas the other 5 actions have probabilistic outcomes:

2 tickets and correct vehicle (60% success):

$$H(S'|A = \text{correct}, C = 2) = - [.6 \times \log_2(.6) + .4 \times \log_2(.4)] = .97 \text{ bits}$$

2 tickets and wrong vehicle (10% success):

$$H(S'|A = \text{wrong}, C = 2) = - [.1 \times \log_2(.1) + .9 \times \log_2(.9)] = .47 \text{ bits}$$

$$H(S'|C = 0-1) = - [.2 \times \log_2(.2) + .8 \times \log_2(.8)] = .72 \text{ bits}$$

Thus the total conditional entropy of the elastic environment is: $H(S'|S, A, C) = (1/7) \times .97 + (1/7) \times .47 + (3/7) \times .72 = .52$ bits

Step 3: Calculating $I(S'|A, S)$

$$\text{Inelastic environment: } I(S'; A, C | S) = H(S'|S) - H(S'|S, A, C) = 1 - 0.1 = .9 \text{ bits}$$

$$\text{Elastic environment: } I(S'; A, C | S) = H(S'|S) - H(S'|S, A, C) = .91 - .52 = .39 \text{ bits}$$

Thus, the inelastic environment offers higher information-theoretic controllability (.9 bits) compared to the elastic environment (.39 bits).

Of note, even if each combination of cost and success/failure to reach the goal is defined as a distinct outcome, then information-theoretic controllability is higher for the inelastic (2.81 bits) than for the elastic (2.30 bits) environment. These calculations are now included in the Supplementary materials (Supplementary Note 1).

In sum, for both definitions of controllability, we see that environments can be more elastic yet less controllable. We have also revised the manuscript to clarify this distinction (lines 21-28):

“While only controllable environments can be elastic, the inverse is not necessarily true – controllability can be high, yet inelastic to invested resources – for example, choosing between bus routes affords equal control over commute time to anyone who can afford the basic fare (Figure 1; Supplementary Note 1). That said, since all actions require some resource investment, no controllable environment is completely inelastic when considering the full spectrum of possible agents, including those with insufficient resources to act (e.g., those unable to purchase a bus fare or pay for a fixed-price meal).”

Reviewer 3 (Public review):

*A bias in how people infer the amount of control they have over their environment is widely believed to be a key component of several mental illnesses including depression, anxiety, and addiction. Accordingly, this bias has been a major focus in computational models of those disorders. However, all of these models treat control as a unidimensional property, roughly, how strongly outcomes depend on action. This paper proposes---correctly, I think---that the intuitive notion of "control" captures multiple dimensions in the relationship between action and outcome is multi-dimensional. In particular, the authors propose that the degree to which outcome depends on how much *effort* we exert, calling this dimension the "elasticity of control". They additionally propose that this dimension (rather than the more holistic notion of controllability) may be specifically impaired in certain types of psychopathology. This idea thus has the potential to change how we think about mental disorders in a substantial way, and could even help us better understand how healthy people navigate challenging decision-making problems.*

Unfortunately, my view is that neither the theoretical nor empirical aspects of the paper really deliver on that promise. In particular, most (perhaps all) of the interesting claims in the paper have weak empirical support.

We appreciate the Reviewer's thoughtful engagement with our research and recognition of the potential significance of distinguishing between different dimensions of control in understanding psychopathology. We believe that all the Reviewer's comments can be addressed with clarifications or additional analyses, as detailed below.

Starting with theory, the elasticity idea does not truly "extend" the standard control model in the way the authors suggest. The reason is that effort is simply one dimension of action. Thus, the proposed model ultimately grounds out in how strongly our outcomes depend on our actions (as in the standard model). Contrary to the authors' claims, the elasticity of control is still a fixed property of the environment. Consistent with this, the computational model proposed here is a learning model of this fixed environmental property. The idea is still valuable, however, because it identifies a key dimension of action (namely, effort) that is particularly relevant to the notion of perceived control. Expressing the elasticity idea in this way might support a more general theoretical formulation of the idea that could be applied in other contexts. See Huys & Dayan (2009), Zorowitz, Momennejad, & Daw (2018), and Gagne & Dayan (2022) for examples of generalizable formulations of perceived control.

We thank the Reviewer for the suggestion that we formalize our concept of elasticity to resource investment, which we agree is a dimension of action. We first note that we have not argued against the claim that elasticity is a fixed property of the environment. We surmise the Reviewer might have misread our statement that “controllability is not a fixed property of the environment”. The latter statement is motivated by the observation that controllability is often higher for agents that can invest more resources (e.g., a richer person can buy more things). We clarify this in our revision of the manuscript in lines 8-15 (changes in bold):

“The degree of control we possess over our environment, however, may itself depend on the resources we are willing and able to invest. For example, the control a biker has over their commute time depends on the power they are willing and able to invest in pedaling. In this respect, a highly trained biker would typically have more control than a novice. Likewise, the control a diner in a restaurant has over their meal may depend on how much money they have to spend. In such situations, controllability is not fixed but rather elastic to available resources (i.e., in the same sense that supply and demand may be elastic to changing prices[14]).”

To formalize elasticity, we build on Huys & Dayan’s definition of controllability¹ as the fraction of reward that is controllably achievable, χ (though using information-theoretic definitions[2,3] would work as well). To the extent that this fraction depends on the amount of resources the agent is able and willing to invest ($\max C$), this formulation can be probabilistically computed without information about the particular agent involved, specifically, by assuming a certain distribution of agents with different amounts of available resources. This would result in a probability distribution over χ . Elasticity can thus be defined as the amount of information obtained about controllability due to knowing the amount of resources available to the agent: $I(\chi; \max C)$. We have added this formal definition to the manuscript (lines 15-20):

“To formalize how elasticity relates to control, we build on an established definition of controllability as the fraction of reward that is controllably achievable[15], χ . Uncertainty about this fraction could result from uncertainty about the amount of resources that the agent is able and willing to invest, $\max C$. Elasticity can thus be defined as the amount of information obtained about controllability by knowing the amount of available resources: $I(\chi; \max C)$.”

Turning to experiment, the authors make two key claims: (1) people infer the elasticity of control, and (2) individual differences in how people make this inference are importantly related to psychopathology. Starting with claim 1, there are three sub-claims here; implicitly, the authors make all three. (1A) People’s behavior is sensitive to differences in elasticity, (1B) people actually represent/track something like elasticity, and (1C) people do so naturally as they go about their daily lives. The results clearly support 1A. However, 1B and 1C are not supported. Starting with 1B, the experiment cannot support the claim that people represent or track elasticity because the effort is the only dimension over which participants can engage in any meaningful decision-making (the other dimension, selecting which destination to visit, simply amounts to selecting the location where you were just told the treasure lies). Thus, any adaptive behavior will necessarily come out in a sensitivity to how outcomes depend on effort. More concretely, any model that captures the fact that you are more likely to succeed in two attempts than one will produce the observed behavior. The null models do not make this basic assumption and thus do not provide a useful comparison.

We appreciate the Reviewer’s critical analysis of our claims regarding elasticity inference, which as detailed below, has led to an important new analysis that strengthens the study’s conclusions. However, we respectfully disagree with two of the Reviewer’s arguments. First, resource investment was not the only meaningful decision dimension in our task, since participant also needed to choose the correct vehicle to get to the right destination. That this was not trivial is evidenced by our exclusion of over 8% of participants who made incorrect vehicle choices more than 10% of the time. Included participants also occasionally erred in this choice (mean error rate = 3%, range [0-10%] now specified in lines 363-366).

Second, the experimental task cannot be solved well by a model that simply tracks how outcomes depend on effort because 20% of the time participants reached the treasure despite

failing to board their vehicle of choice. In such cases, reward outcomes and control were decoupled. Participants could identify when this was the case by observing the starting location (since depending on the starting location, the treasure location could have been automatically reached by walking), which was revealed together with the outcome. To determine whether participants distinguished between control-related and non-control-related reward, we have now fitted a variant of our model to the data that allows learning from each of these kinds of outcomes by means of a different free parameter. The results show that participants learned considerably more from control-related outcomes. They were thus not merely tracking outcomes, but specifically inferred when outcomes can be attributed to control. We now include this new analysis in the revised manuscript (Methods lines 648-661):

“To ascertain that participants were truly learning latent estimates of controllability rather than simpler associations, we conducted two complementary analyses.

First, we implemented a simple Q-learning model that directly maps ticket quantities to expected values based on reward prediction errors, without representing latent controllability. This associative model performed substantially worse than even our simple controllability model (log Bayes Factor ≥ 1854 on the combined datasets). Second, we fitted a variant of the elastic controllability model that compared learning from control-related versus chance outcomes via separate parameters (instead of assuming no learning from chance outcomes). Chance outcomes were observed by participants in the 20% of trials where reward and control were decoupled, in the sense that participants reached the treasure regardless of whether they boarded their vehicle of choice. Results showed that participants learned considerably more from control-related, as compared to chance, outcomes (mean learning ratio=1.90, CI= [1.83, 1.97]). Together, these analyses show that participants were forming latent controllability estimates rather than direct action-outcome associations.”

Controllability inference by itself, however, still does not suffice to explain the observed behavior. This is shown by our ‘controllability’ model, which learns to invest more resources to improve control, yet still fails to capture key features of participants’ behavior, as detailed in the manuscript. This means that explaining participants’ behavior requires a model that not only infers controllability—beyond merely outcome probability—but also assumes a priori that increased effort could enhance control. Building these a priori assumption into the model amounts to embedding within it an understanding of elasticity – the idea that control over the environment may be increased by greater resource investment.

That being said, we acknowledge the value in considering alternative computational formulations of adaptation to elasticity, as now expressed in the revised discussion (lines 326-333; reproduced below in response to the Reviewer’s comment on updating controllability beliefs when losing with less than 3 tickets).

For 1C, the claim that people infer elasticity outside of the experimental task cannot be supported because the authors explicitly tell people about the two notions of control as part of the training phase: “To reinforce participants’ understanding of how elasticity and controllability were manifested in each planet, [participants] were informed of the planet type they had visited after every 15 trips.” (line 384).

We thank the Reviewer for highlighting this point. We agree that our experimental design does not test whether people infer elasticity spontaneously. However, our research question was whether people can distinguish between elastic and inelastic controllability. The results strongly support that they can, and this does have potential implications for behavior outside of the experimental task. Specifically, to the extent that people are aware that in some contexts additional resource investment improves control, whereas in other contexts it does not, then our results indicate that they would be able to distinguish between these two kinds

of contexts through trial-and-error learning. That said, we agree that investigating whether and how people spontaneously infer elasticity is an interesting direction for future work. We have now added this to the discussion of future directions (lines 287-295):

“Additionally, real life typically doesn’t offer the streamlined recurrence of homogenized experiences that makes learning easier in experimental tasks, nor are people systematically instructed and trained about elastic and inelastic control in each environment. These complexities introduce substantial additional uncertainty into inferences of elasticity in naturalistic settings, thus allowing more room for prior biases to exert their influences. The elasticity biases observed in the present studies are therefore likely to be amplified in real-life behavior. Future research should examine how these complexities affect judgments about the elasticity of control to better understand how people allocate resources in real-life.”

Finally, I turn to claim 2, that individual differences in how people infer elasticity are importantly related to psychopathology. There is much to say about the decision to treat psychopathology as a unidimensional construct. However, I will keep it concrete and simply note that CCA (by design) obscures the relationship between any two variables. Thus, as suggestive as Figure 6B is, we cannot conclude that there is a strong relationship between Sense of Agency and the elasticity bias---this result is consistent with any possible relationship (even a negative one). The fact that the direct relationship between these two variables is not shown or reported leads me to infer that they do not have a significant or strong relationship in the data.

We agree that CCA is not designed to reveal the relationship between any two variables. However, the advantage of this analysis is that it pulls together information from multiple variables. Doing so does not treat psychopathology as unidimensional. Rather, it seeks a particular dimension that most strongly correlates with different aspects of task performance.

This is especially useful for multidimensional psychopathology data because such data are often dominated by strong correlations between dimensions, whereas the research seeks to explain the distinctions between the dimensions. Similar considerations apply to the multidimensional task parameters, which although less correlated, may still jointly predict the relevant psychopathological profile better than each parameter does in isolation. Thus, the CCA enabled us to identify a general relationship between task performance and psychopathology that accounts for different symptom measures and aspects of controllability inference.

Using CCA can thus reveal relationships that do not readily show up in two-variable analyses. Indeed, the direct correlation between Sense of Agency (SOA) and elasticity bias was not significant – a result that, for completeness, we now report in Supplementary Figure 3 along with all other direct correlations. We note, however, that the CCA analysis was preregistered and its results were replicated. Additionally, participants scoring higher on the psychopathology profile also overinvested resources in inelastic environments but did not futilely invest in uncontrollable environments (Figure 6A), providing external validation to the conclusion that the CCA captured meaningful variance specific to elasticity inference. Most importantly, an auxiliary analysis specifically confirmed the contributions of both elasticity bias (Figure 6D, middle plot) and, although not reported in the original paper, of the Sense of Agency score (SOA; $p=.03$ permutation test; see updated Figure 6D, bottom plot) to the observed canonical correlation. The results thus enable us to safely conclude that differences in elasticity inferences are significantly associated with a profile of control-related psychopathology to which SOA contributed significantly. We now report this when presenting the CCA results (lines 255-257):

“Loadings on the side of psychopathology were dominated by an impaired sense of agency (SOA; contribution to canonical correlation: $p=.03$, Figure 6D, bottom plot), along with obsessive compulsive symptoms (OCD), and social anxiety (LSAS) – all symptoms that have been linked to an impaired sense of control[22-25].”

Finally, whereas interpretation of individual CCA loadings that were not specifically tested remains speculative, we note that the pattern of loadings largely replicated across the initial and replication studies (see Figure 6B), and aligns with prior findings. For instance, the positive loadings of SOA and OCD match prior suggestions that a lower sense of control leads to greater compensatory effort⁷, whereas the negative loading for depression scores matches prior work showing reduced resource investment in depression[5-6].

We have now revised the manuscript to clarify the justification for our analytical approach (lines 236-248):

“To examine whether the individual biases in controllability and elasticity inference have psychopathological ramifications, we assayed participants on a range of self-report measures of psychopathologies previously linked to a distorted sense of control (see Methods, pg. 24). Examining the direct correlations between model parameters and psychopathology measures (reported in Supplementary Figure 3) does not account for the substantial variance that is typically shared among different forms of psychopathology. For this reason, we instead used a canonical correlation analysis (CCA) to identify particular dimensions within the parameter and psychopathology spaces that most strongly correlate with one another.”

We also now include a cautionary note in the discussion (lines 309-315):

“Whereas our pre-registered CCA effectively identified associations between task parameters and a psychopathological profile, this analysis method does not directly reveal relationships between individual variables. Auxiliary analyses confirmed significant contributions of both elasticity bias and sense of agency to the observed canonical correlation, but the contribution of other measures remains to be determined by future work. Such work could employ other established measures of agency, including both behavioral indices and subjective self-reports, to better understand how these constructs relate across different contexts and populations.”

*There is also a feature of the task that limits our ability to draw strong conclusions about individual differences in elasticity inference. As the authors clearly acknowledge, the task was designed "to be especially sensitive to overestimation of elasticity" (line 287). A straightforward consequence of this is that the resulting *empirical* estimate of estimation bias (i.e., the $\gamma_{\text{elasticity}}$ parameter) is itself biased. This immediately undermines any claim that references the directionality of the elasticity bias (e.g. in the abstract). Concretely, an undirected deficit such as slower learning of elasticity would appear as a directed overestimation bias. When we further consider that elasticity inference is the only meaningful learning/decisionmaking problem in the task (argued above), the situation becomes much worse. Many general deficits in learning or decision-making would be captured by the elasticity bias parameter. Thus, a conservative interpretation of the results is simply that psychopathology is associated with impaired learning and decision-making.*

We apologize for our imprecise statement that the task was ‘especially sensitive to overestimation of elasticity’, which justifiably led to Reviewer’s concern that slower elasticity learning can be mistaken for elasticity bias. To make sure this was not the case, we made use of the fact that our computational model explicitly separates bias direction (λ) from the rate of learning ϕ through two distinct parameters, which initialize the prior concentration and

mean of the model's initial beliefs concerning elasticity (see Methods pg. 23). The higher the concentration of the initial beliefs (ϵ), the slower the learning. Parameter recovery tests confirmed that our task enables acceptable recovery of both the bias $\lambda_{\text{elasticity}}$ ($r=.81$) and the concentration $\epsilon_{\text{elasticity}}$ ($r=.59$) parameters. And importantly, the level of confusion between the parameters was low (confusion of 0.15 for $\epsilon_{\text{elasticity}} \rightarrow \lambda_{\text{elasticity}}$ and 0.04 for $\lambda_{\text{elasticity}} \rightarrow \epsilon_{\text{elasticity}}$). This result confirms that our task enables dissociating elasticity biases from the rate of elasticity learning.

Moreover, to validate that the minimal level of confusion existing between bias and the rate of learning did not drive our psychopathology results, we re-ran the CCA while separating concentration from bias parameters. The results (figure below) demonstrate that differences in learning rate (ϵ) had virtually no contribution to our CCA results, whereas the contribution of the pure bias (λ) was preserved.

We now report on this additional analysis in the text (lines 617-627):

“To capture prior biases that planets are controllable and elastic, we introduced parameters $\lambda_{\text{controllability}}$ and $\lambda_{\text{elasticity}}$, each computed by multiplying the direction ($\lambda - 0.5$) and strength (ϵ) of individuals' prior belief. $\epsilon_{\text{controllability}}$ and $\epsilon_{\text{elasticity}}$ range between 0 and 1, with values above 0.5 indicating a bias towards high controllability or elasticity, and values below 0.5 indicating a bias towards low controllability or elasticity. $\epsilon_{\text{controllability}}$ and $\epsilon_{\text{elasticity}}$ are positively valued parameters capturing confidence in the bias. Parameter recovery analyses confirmed both good recoverability (see S2 Table) and low confusion between bias direction and strength ($\epsilon_{\text{controllability}} \rightarrow \lambda_{\text{controllability}} = -.07$, $\lambda_{\text{controllability}} \rightarrow \epsilon_{\text{controllability}} = .16$, $\epsilon_{\text{elasticity}} \rightarrow \lambda_{\text{elasticity}} = .15$, $\lambda_{\text{elasticity}} \rightarrow \epsilon_{\text{elasticity}} = .04$), ensuring that observed biases and their relation to psychopathology do not merely reflect slower learning (Supplementary Figure 4), which can result from changes in bias strength but not direction.”

We also more precisely articulate the impact of providing participants with three free tickets at their initial visits to each planet.

Showing that a model parameter correlates with the data it was fit to does not provide any new information, and cannot support claims like "a prior assumption that control is likely available was reflected in a futile investment of resources in uncontrollable environments." To make that claim, one must collect independent measures of the assumption and the investment.

We apologize if this and related statements seemed to be describing independent findings. They were meant to describe the relationship between model parameters and model-independent measures of task performance. It is inaccurate, though, to say that they provide no new information, since results could have been otherwise. For instance, whether a higher controllability bias maps onto resource misallocation in uncontrollable environments (as we observed) depends on the range of this parameter in our population sample. Had the range been more negative, a higher controllability bias could have instead manifested as optimal allocation in controllable environments. Additionally, these analyses serve two other purposes: as a validity check, confirming that our computational model effectively captured observed individual differences, and as a help for readers to understand what each parameter in our model represents in terms of observable behavior. We now better clarify the descriptive purposes of these regressions (lines 214-220, 231-235):

“To clarify how fitted model parameters related to observable behavior, we regressed participants' opt-in rates and extra ticket purchases on the parameters (Figure 6A) ...”

“... In sum, the model parameters captured meaningful individual differences in how participants allocated their resources across environments, with the controllability parameter primarily explaining variance in resource allocation in uncontrollable

environments, and the elasticity parameter primarily explaining variance in resource allocation in environments where control was inelastic.”

Did participants always make two attempts when purchasing tickets? This seems to violate the intuitive model, in which you would sometimes succeed on the first jump. If so, why was this choice made? Relatedly, it is not clear to me after a close reading how the outcome of each trial was actually determined.

We thank the Reviewer for highlighting the need to clarify these aspects of the task in the revised manuscript.

When participants purchased two extra tickets, they attempted both jumps, and were never informed about whether either of them succeeded. Instead, after choosing a vehicle and attempting both jumps, participants were notified where they arrived at. This outcome was determined based on the cumulative probability of either of the two jumps succeeding. Success meant that participants arrived at where their chosen vehicle goes, whereas failure meant they walked to the nearest location (as determined by where they started from).

Though it is unintuitive to attempt a second jump before seeing whether the first succeed, this design choice ensured two key objectives. First, that participants would consistently need to invest not only more money but also more effort and time in planets with high elastic controllability. Second, that the task could potentially generalize to the many real-world situations where the amount of invested effort has to be determined prior to seeing any outcome, for instance, preparing for an exam or a job interview. We now explicitly state these details when describing the experimental task (lines 393-395):

“When participants purchased multiple tickets, they made all boarding attempts in sequence without intermediate feedback, only learning whether they successfully boarded upon reaching their final destination. This served two purposes. First, to ensure that participants would consistently need to invest not only more money but also more effort and time in planets with high elastic controllability. Second, to ensure that results could potentially generalize to the many real-world situations where the amount of invested effort has to be determined prior to seeing any outcome (e.g., preparing for an exam or a job interview).”

It should be noted that the model is heuristically defined and does not reflect Bayesian updating. In particular, it overestimates control by not using losses with less than 3 tickets (intuitively, the inference here depends on your beliefs about elasticity). I wonder if the forced three-ticket trials in the task might be historically related to this modeling choice.

We apologize for not making this clear, but in fact losing with less than 3 tickets does reduce the model’s estimate of available control. It does so by increasing the elasticity estimates ($a_{\text{elastic} \geq 1}, a_{\text{elastic} \geq 2}$ parameters), signifying that more tickets are needed to obtain the maximum available level of control, thereby reducing the average controllability estimate across ticket investment options. We note this now in the presentation of the computational model (caption Figure 4):

“A failure to board does not change estimated maximum controllability, but rather suggests that 1 ticket might not suffice to obtain control ($a_{\text{elastic} \geq 1} + 1$; *light green diminished*). As a result, the model’s estimate of average controllability across ticket options is reduced.”

It would be interesting to further develop the model such that losing with less than 3 tickets would also impact inferences concerning the maximum available control, depending on present beliefs concerning elasticity, but the forced three-ticket purchases already expose participants to the maximum available control, and thus, the present data may not be best suited to test such a model. These trials were implemented to minimize individual differences

concerning inferences of maximum available control, thereby focusing differences on elasticity inferences. We now explicitly address these considerations in the revised discussion (lines 326-333) with the following:

“Future research could explore alternative models for implementing elasticity inference that extend beyond our current paradigm. First, further investigation is warranted concerning how uncertainty about controllability and its elasticity interact. In the present study, we minimized individual differences in the estimation of maximum available control by providing participants with three free tickets at their initial visits to each planet. We made this design choice to isolate differences in the estimation of elasticity, as opposed to maximum controllability. To study how these two types of estimations interact, future work could benefit from modifying this aspect of our experimental design.”

Furthermore, we have now tested a Bayesian model suggested by Reviewer 1, but we found that this model fitted participants' choices worse (see details in the response to Reviewer 1's comments).

Recommendations for the authors:

Reviewer 1 (Recommendations for the authors):

In the introduction, the definition of controllability and elasticity, and the scope of "resources" investigated in the current study were unclear. If I understand correctly, controllability is defined as "the degree to which actions influence the probability of obtaining a reward", and elasticity is defined as the change in controllability based on invested resources. This would define the controllability of the environment and the elasticity of controllability of the environment. However, phrases such as "elastic environment" seem to imply that elasticity can directly attach to an environment, instead of attaching to the controllability of the environment.

We thank the Reviewer for highlighting the need to clarify our conceptualization of elasticity and controllability. We now provide formal definitions of both, with controllability defined as the fraction of controllably achievable reward[1], and elasticity as the reduction in uncertainty about controllability due to knowing the amount of resources the agent is willing and able to invest (see further details in the response to Reviewer 3's public comments). In the revised manuscript, we now use more precise language to clarify that elasticity is a property of controllability, not of environments themselves. In addition, we now clarify that the current study manipulated monetary, attentional effort, and time costs together (see further details in the response to Reviewer 1's public comments).

(2) Some of the real-world examples were confusing. For example, the authors mention that investing additional effort due to the belief that this leads to better outcomes in OCD patients is overestimated elasticity, but exercising due to the belief that this can make one taller is overestimated controllability. What's the distinction between the examples? The example of the chess expert practicing to win against a novice, because the amount of effort they invest would not change their level of control over the outcome is also unclear. If the control over the outcome depends on their skill set, wouldn't practicing influence the control over the outcome? In the case of the meeting time example, wouldn't the bus routes differ in their time investments even though they are the same price? In addition to focusing the introductory examples around monetary resources, I would also generally recommend tightening the link between those examples and the experimental task.

We thank the Reviewer for highlighting the need to clarify the examples used to illustrate elasticity and controllability. We have now revised these examples to more clearly distinguish

between the concepts and to strengthen their connection to the experimental task.

Regarding the OCD example, the possibility that OCD patients overestimate elasticity comes from research suggesting they experience low perceived control but nevertheless engage in excessive resource investment², reflecting a belief that only through repeated and intense effort can they achieve sufficient control over outcomes. As an example, consider an OCD patient investing unnecessary effort in repeatedly locking their door. This behavior cannot result from an overestimation of controllability because controllability truly is close to maximal. It also cannot result from an underestimation of the maximum attainable control, since in that case investing more effort is futile. Such behavior, however, can result from an overestimation of the degree to which controllability requires effort (i.e., overestimation of elasticity).

Similarly, with regards to the chess expert, we intended to illustrate a situation where given their current level, the chess expert is already virtually guaranteed to win, such that additional practice time does not improve their chances. Conversely, the height example illustrates overestimated controllability because the outcome (becoming taller through exercise) is in fact not amenable to control through any amount of resource investment.

Finally, the meeting time example was meant to illustrate that if the desired outcome is reaching a meeting in time, then different bus routes that cost the same provide equal control over this outcome to anyone who can afford the basic fare. This demonstrates inelastic controllability with respect to money, as spending more on transportation doesn't increase the probability of reaching the meeting on time. The Reviewer correctly notes that time investment may differ between routes. However, investing more time does not improve the expected outcome. This illustrates that inelastic controllability does not preclude agents from investing more resources, but such investment does not increase the fraction of controllably achievable reward (i.e., the probability of reaching the meeting in time).

In the revised manuscript, we've refined each of the above examples to better clarify the specific resources being considered, the outcomes they influence, and their precise relationship to both elasticity and controllability:

OCD (lines 40-43): Conversely, the repetitive and unusual amount of effort invested by people with obsessive-compulsive disorder in attempts to exert control^[23,24] could indicate an overestimation of elasticity, that is, a belief that adequate control can only be achieved through excessive and repeated resource investment^[25].

Chess expert (54-57): Alternatively, they may do so because they overestimate the elasticity of control – for example, a chess expert practicing unnecessarily hard to win against a novice, when their existing skill level already ensures control over the match's outcome.

Height (lines 53-54): A given individual, for instance, may tend to overinvest resources because they overestimate controllability – for example, exercising due to a misguided belief that that this can make one taller, when in fact height cannot be controlled.

Meeting time (lines 26-28): Choosing between bus routes affords equal control over commute time to anyone who can afford the basic fare (Figure 1).

Methods

(1) In the elastic controllability model definition, controllability is defined as "the belief that boarding is possible" (with any number of tickets). The definition again is different from in the task description where controllability is defined as "the probability of the chosen vehicle stopping at the platform if purchasing a single ticket."

We clarify that "the probability of the chosen vehicle stopping at the platform if purchasing a single ticket" is our definition for inelastic controllability, as opposed to overall/maximum controllability, as stated here (lines 101-103):

"We defined inelastic controllability as the probability that even one ticket would lead to successfully boarding the vehicle, and elastic controllability as the degree to which two extra tickets would increase that probability."

Overall controllability is the summation of the two. This summation is referred to in the elastic controllability model definition as the "the belief that boarding is possible". We now clarify this in the caption to figure 4:

Elastic Controllability model: Represents beliefs about maximum controllability (black outline) and the degree to which one or two extra tickets are necessary to obtain it. These beliefs are used to calculate the expected control when purchasing 1 ticket (inelastic controllability) and the additional control afforded by 2 and 3 tickets (elastic controllability).

We also clarify this in the methods when describing the parameterization of the model (lines 529-531):

The expected value of one beta distribution (defined by $a_{\text{sub} > \text{control}}$, $b_{\text{sub} > \text{control}}$) represents the belief that boarding is possible (controllability) with any number of tickets.

(2) The free parameter K is confusing. What is the psychological meaning of this parameter? Is it there just to account for the fact that failure with 3 tickets made participants favor 3 tickets or is there meaning attached to including this parameter?

This parameter captures how participants update their beliefs about resource requirements after failing to board with maximum resource investment. Our psychological interpretation is that participants who experience failure despite maximum investment (3 tickets) prioritize resolving uncertainty about whether control is fundamentally possible (before exploring whether control is elastic), which can only be determined by continuing to invest maximum resources.

We now clarify this in the methods (lines 555-559):

To account for our finding that failure with 3 tickets made participants favor 3, over 1 and 2, tickets, we introduced a modified elastic controllability* model, wherein purchasing extra tickets is also favored upon receiving evidence of low controllability (loss with 3 tickets). This effect was modulated by a free parameter κ which reflects a tendency to prioritize resolving uncertainty about whether control is at all possible by investing maximum resources.

This interpretation is supported by our analysis of 3-ticket choice trajectories (Supplementary Figure 2 presented in response to Reviewer 2). As shown in the figure, participants who win less than 50% of their 3-ticket attempts persistently purchase 3 tickets over the first 10 trials, despite frequent failures. This persistence gradually declines as participants accumulate evidence about their limited control, corresponding with an increase in opt-out rates.

(3) Some additional details about the task design would be helpful. It seems that participants first completed 90 practice trials and were informed of the planet type every 15 trials (6 times during practice). What message is given to the participants about the planets? Did the authors analyze the last 15 trials of each condition in the regression analysis, and all 30 trials in the modeling analysis? How does the computational model (especially the prior beliefs parameters) reset when the planet changes? How do points accumulate over the session and/or are participants motivated to budget the points? Is it

possible for participants to accumulate many points and then switch to a heuristic of purchasing 3 tickets on each trial?

We apologize for not previously clarifying these details of the experimental design.

During practice blocks, participants received explicit feedback about each planet's controllability characteristics, to help them understand when additional resources would or would not improve their boarding success. For high inelastic controllability planets, the message read: "Your ride actually would stop for you with 1 ticket! So purchasing extra tickets, since they do cost money, is a WASTE." For low controllability planets: "Doesn't seem like the vehicle stops for you nor does purchasing extra tickets help." Lastly, for high elastic controllability planets: "Hopefully by now it's clear that only by purchasing 3 tickets (LOADING AREA) are you consistently successful in catching your ride." We now include these messages in the methods section describing the task (lines 453-458).

We indeed analyzed the last 15 trials of each condition in the regression analysis, and all 30 trials in the modeling analysis. Whereas the modeling attempted to explain participants' learning process, the regression focused on explaining the resultant behavior, which in our pilot data (N=19), manifested fairly stably in the last 15 trials (ticket choices SD = 0.33 compared to .63 in the first 15 trials). The former is already stated in the text (lines 409-415), and we now also clarify the latter when discussing the model fitting procedure (line 695):

Reinforcement-learning models were fitted to all choices made by participants via an expectation maximization approach used in previous work.

The computational model was initialized with the same prior parameters for all planets. When a participant moved to a new planet, the model's beliefs were reset to these prior values, capturing how participants would approach each new environment with their characteristic expectations about controllability and elasticity. We now clarify this in the methods (line 628):

For each new planet participants encountered, these parameters were used to initialize the beta distributions representing participants' beliefs

Points accumulated across all planets throughout the session, with participants explicitly motivated to maximize their total points as this directly determined their monetary bonus payment. To address the Reviewer's question about changes in ticket purchasing behavior, we conducted a mixed probit regression examining whether accumulated points influenced participants' decisions to purchase extra tickets. We did not find such an effect ($\beta_{\text{coins accumulated}} = .01$ $p = .87$), indicating that participants did not switch to simple heuristic strategies after accumulating enough coins. We now report this analysis in the methods (lines 421-427):

Points accumulated across all planets throughout the session, with participants explicitly motivated to maximize their total points as this directly determined their monetary bonus payment. To ensure that accumulated gains did not lead participants to adopt a simple heuristic strategy of always purchasing 3 tickets, we conducted a mixed probit regression examining whether the number of accumulated coins influenced participants' decisions to purchase extra tickets. We did not find such an effect ($\beta_{\text{coins accumulated}} = .01$ $p = .87$), ruling out the potential strategy shift.

Following the modeling section, it may be helpful to have a table of the fitted models, the parameters of each model, and the meaning/interpretation of each parameter.

We thank the Reviewer for this suggestion. We have now added a table (Supplementary Table 3) that summarizes all fitted models, their parameters, and the meaning/interpretation of

each parameter.

(1) The conclusions from regressing the task choices (opt-in rates and ticket purchases) on the fitted parameters seem confusing given that the model parameters were fitted on the task behavior, and the relationship between these variables seems circular. For example, the authors found that preferences for purchasing 2 or 3 tickets (α_2 and α_3 ; computational parameters) were associated with purchasing more tickets (task behavior). But wouldn't this type of task behavior be what the parameters are explaining? It's not clear whether these correlation analyses are about how individuals allocate their resources or about the validity check of the parameters. Perhaps analyses on individual deviation from the optimal strategy and parameter associations with such deviation are better suited for the questions about whether individual biases lead to resource misallocation.

We thank the Reviewer for highlighting this seeming confusion. These regressions were meant to describe the relationship between model parameters and model-independent measures of task performance. This serves three purposes. First, a validity check, confirming that our computational model effectively captured observed individual differences. Second, to help readers understand what each parameter in our model represents in terms of observable behavior. Third, to examine in greater detail how parameter values specifically mapped onto observable behavior. For instance, whether a higher controllability bias maps onto resource misallocation in uncontrollable environments (as we observed) depends on the range of this parameter in our population sample. Had the range been more negative, a higher controllability bias could have instead manifested as optimal allocation in controllable environments. We now better clarify the descriptive purposes of these regressions (lines 214-220, 231-235):

To clarify how fitted model parameters related to observable behavior, we regressed participants' opt-in rates and extra ticket purchases on the parameters (Figure 6A) ...

... In sum, the model parameters captured meaningful individual differences in how participants allocated their resources across environments, with the controllability parameter primarily explaining variance in resource allocation in uncontrollable environments, and the elasticity parameter primarily explaining variance in resource allocation in environments where control was inelastic.

Regarding the suggestion to analyze deviation from optimal strategy, this corresponds with our present approach in that opting in is always optimal in high controllability environments and always non-optimal in low controllability environments, and similarly, purchasing extra tickets is always optimal in elastic controllability environments and always non-optimal elsewhere. Thus, positive or negative coefficients can be directly translated into closer or farther from optimal, depending on the planet type, as indicated in the figure by color. We now clarify this mapping in the figure legend:

(2) Minor: The legend of Figure 6A is difficult to read. It might be helpful to label the colors as their planet types (low controllability, high elastic controllability, high inelastic controllability).

We thank the Reviewer for this helpful suggestion. We have revised the figure accordingly.

Reviewer 2 (Recommendations for the authors):

As noted above, I'm not sure I agree with (or perhaps don't fully understand) the claims the authors make about the distinctions between their "elastic" and "inelastic" experimental conditions. Let's take the travel example from Figure 1 - is this not just an

example of "hierarchical" controllability calculations? In other words, in the elastic example, my choice is between going one speed or another (i.e., exerting more or less effort), and in the inelastic example, my choice is first, which route to take (also a consideration of speed, but with lower effort costs than the elastic scenario), and second, an estimate of the time cost (not within my direct control, but could be estimated). In the elastic scenarios, additional value considerations vary between options, and in others (inelastic), they don't, with control over the first choice point (which bus route to choose, or which lunch option to take), but not over the price. I wonder if the paper would be better framed (or emphasized) as exploring the influences of effort and related "costs" of control. There isn't really such a thing as controllability that does not have any costs associated with it (whether that be action costs, effort, money, or simply scenario complexity).

We thank the Reviewer for highlighting the need to clarify our distinction between elastic and inelastic controllability as it manifests in our examples. We first clarify that elasticity concerns how controllability varies with resources, not costs. Though resource investment and costs are often tightly linked, that is not always the case, especially not when comparing between agents. For example, it may be equally difficult (i.e., costly) for a professional biker to pedal at a high speed as it is for a novice to pedal at a medium speed, simply because the biker's muscles are better trained. This resource advantage increases the biker's control over his commute time without incurring additional costs as compared to the novice. We now clarify this distinction in the text by revising our example to (lines 9-11):

"For example, the control a biker has over their commute time depends on the power they are willing and able to invest in pedaling. In this respect, a highly trained biker would typically have more control than a novice."

Second, whereas in our examples additional value considerations indeed vary in elastic environments, that does not have to be the case, and indeed, that is not the case in our experiment. In our experimental task, participants are given the option to purchase as many tickets as they wish regardless of whether they are in an elastic or an inelastic environment.

We agree that elastic environments often raise considerations regarding the cost of control (for instance, whether it is worth it to pedal harder to get to the destination in time). To consider this cost against potential payoffs, however, the agent must first determine what are the potential payoffs – that is, it must determine the degree to which controllability is elastic to invested resources. It is this antecedent inference that our experiment studies. We uniquely study this inference using environments where control may not only be low or high, but also, where high control may or may not require additional resource investments. We now clarify this point in Figure 1's caption:

"In all situations, agents must infer the degree to which controllability is elastic to be able to determine whether the potential gains in control outweigh the costs of investing additional resources (e.g., physical exertion, money spent, time invested)."

For a formal definition of the elasticity of control, see our response to Reviewer 3's public comments.

Relatedly, another issue I have with the distinctions between inelastic/elastic is that a high/elastic condition has inherently 'more' controllability than a high/inelastic condition, no matter what. For example, in the lunch option scenario, I always have more control in the elastic situation because I have two opportunities to exert choice (food option 'and' cost). Is there really a significant difference, then, between calling these distinctions "elastic/inelastic" vs. "higher/lower controllability?" Not that it's uninteresting to test behavioral differences between these two types of scenarios, just that it seems unnecessary to refer to these as conceptually distinct.

As noted in the response above, control over costs may be higher in elastic environments, but it does not have to be so, as exemplified by the elastic environments in our experimental task. For a fuller explanation of why higher elasticity does not imply higher controllability, see our response to Reviewer 2's public comments.

I also wonder whether it's actually the case that people purchased more tickets in the high control elastic condition simply because this is the optimal solution to achieve the desired outcome, not due to a preference for elastic control. To test this, you would need to include a condition in which people opted to spend more money/effort to have high elastic control in an instance where it was not beneficial to do so.

We appreciate the Reviewer's question about potential preferences for elastic control. We first clarify that participants did not choose which environment type they encountered, so if control was low or inelastic, investing extra resources did not give them more control. Furthermore, our results show that the average participant did not prefer a priori to purchase more tickets. This is evidenced by participants' successful adaptation to inelastic environments wherein they purchased significantly fewer tickets (see Figure 2B and 2C), and by participants' parameter fits, which reveal an a priori bias to assume that controllability is inelastic ($\lambda_{\text{elasticity}} = .16 \pm .19$), as well as a fixed preference against purchasing the full number of tickets ($\alpha_3 = -.74 \pm .37$).

We now clarify these findings by including a table of all parameter fits in the revised manuscript (see response to Reviewer 1).

It was interesting that the authors found that failure with 3 tickets made people more likely to continue to try 3 tickets, however, there is another possible interpretation. Could it be that this is simply evidence of a general controllability bias, where people just think that it is expected that you should be able to exert more money/effort/time to gain control, and if this initially fails, it is an unusual outcome, and they should try again? Did you look at this trajectory over time? i.e., whether repeated tries with 3 tickets immediately followed a failure with 3 tickets? Relatedly, does the perseverance parameter from the model also correlate with psychopathology?

We thank the Reviewer for this suggestion. Our model accounts for a general controllability bias through the $\lambda_{\text{controllability}}$ parameter, which represents a prior belief that planets are controllable. It also accounts, through the $\lambda_{\text{elasticity}}$ parameter, for the prior belief that you should be able to exert more money/effort/time to gain control. Now, our addition of κ to the model captures the observation that failures with 3 tickets made participants more likely to purchase 3 tickets when they opted in. If this observation was due to participants not accepting that the planet is not controllable, then we would expect the increase in 3-ticket purchases when opting in to be coupled with a diminished reduction in opting in. To determine whether this was the case, we tested a variant of our model where κ not only increases the elasticity estimate but also reduces the controllability update (using $\beta_{\text{control}} + (1 - \kappa)$ instead of $\beta_{\text{control}} + 1$) after failures with 3 tickets. However, implementing this coupling diminished the model's fit to the data, as compared to allowing both effects to occur independently, indicating that the increase in 3 ticket purchases upon failing with 3 tickets did not result from participants not accepting that controllability is in fact low. Thus, we maintain our original interpretation that failure with 3 tickets increases uncertainty about whether control is possible at all, leading participants who continue to opt in to invest maximum resources to resolve this uncertainty. We now report these results in the revised text (lines 662-674).

The trajectory over time is consistent this interpretation (new Supplementary Figure 2 shown below). Specifically, we see that under low controllability (0-50%, orange line), over the first

10 trials participants show higher persistence with 3 tickets after failing, despite experiencing frequent failures, but also a higher opt-out probability. As these participants accumulate evidence about their limited control, we observe a gradual decrease in 3-ticket selections that corresponds directly with a further increase in opting out (right panel, orange line). This pattern qualitatively corresponds with the behavior of our computational model (empty circles). We present the results of the new analysis in lines 180-190:

“In fact, failure with 3 tickets even made participants favor 3, over 1 and 2, tickets. This favoring of 3 tickets continued until participants accumulated sufficient evidence about their limited control to opt out (Supplementary Figure 2). Presumably, the initial failures with 3 tickets resulted in an increased uncertainty about whether it is at all possible to control one’s destination. Consequently, participants who nevertheless opted in invested maximum resources to resolve this uncertainty before exploring whether control is elastic.”

Regarding correlations between the perseveration parameter and psychopathology, we have now conducted a comprehensive exploratory analysis of all two-way relationships between parameters and psychopathology scores (new Supplementary Figure 3). Whereas we observed modest negative correlations with social anxiety (LSAS, $r=-0.13$), cyclothymic temperament ($r=0.13$), and alcohol use (AUDIT, $r=-0.13$), none reached statistical significance after FDR correction for multiple comparisons.

Regarding the modeling, I also wondered whether a better alternative model than the controllability model would be a simple associative learning model, where a number of tickets are mapped to outcomes, regardless of elasticity.

We thank the Reviewer for suggesting this alternative model. Following this suggestion, we implemented a simple associative learning model that directly maps each option to its expected value, without a latent representation of elasticity or controllability. Unlike our controllability model which learns the probability of reaching the goal state for each ticket quantity, this associative learning model simply updates option values based on reward prediction errors.

We found that this simple Q-learning model performed worse than even the controllability model at explaining participant data (log Bayes Factor ≥ 1854 on the combined datasets), further supporting our hypothesis that participants are learning latent estimates of control rather than simply associating options with outcomes. We present the results of this analysis in lines 662-664:

We implemented a simple Q-learning model that directly maps ticket quantities to expected values based on reward prediction errors, without representing latent controllability. This associative model performed substantially worse than even our simple controllability model (log Bayes Factor ≥ 1854 on the combined datasets).

Reviewer 3 (Recommendations for the authors):

Please make all materials available, including code (analysis and experiment) and data. Please also provide a link to the task or a video of a few trials of the main task.

We thank the reviewer for this important suggestion. All requested materials are now available at https://github.com/lsolomyak/human_inference_of_elastic_control. This includes all experiment code, analysis code, processed data, and a video showing multiple sample trials of the main task.

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